

TR-1um

Design Rules Summary

for Drawing Layers

2026/07/27

Reflected GitHub issues

<https://github.com/OpenSUSI/TR-1um/issues>

Drawing Layers Table

IMPORTANT

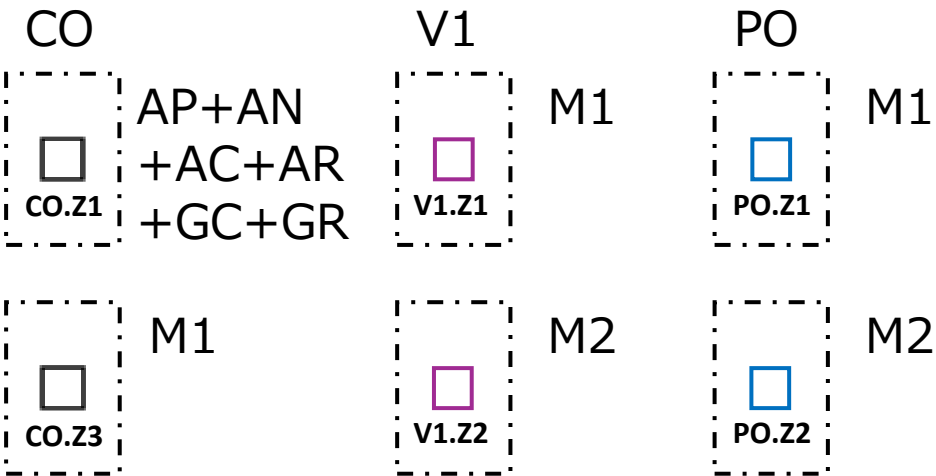
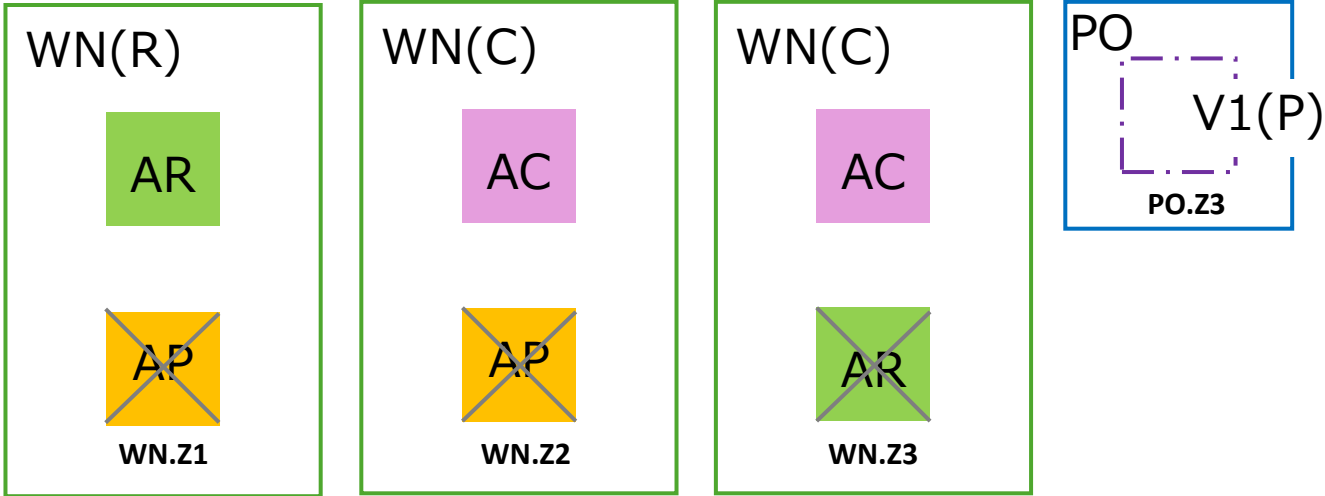
DBU: 0.001um

Scale Factor: 1.0

Min Grid: 0.05um

	Original MASK layers				Proposed Drawing layers					
	Name	NUM	TYPE		Name	NUM	TYPE			
1	PSUB	140	0		WN	140	0			
2	NW	36	0					MDP generatioed		
3	HVNW	141	0					MDP generatioed		
4	L	3	0					MDP generatioed		
					AP	3	1	Active for P+		
					AN	3	2	Active for N+		
					AR	3	3	Active for resistor		
					AC	3	4	Active for capacitance		
5	NF	25	0					MDP generatioed		
6	PF	26	0					MDP generatioed		
7	CL	143	0					MDP generatioed		
8	HPBE	144	0					No USE		
9	HNBE	145	0					No USE		
10	PBE	146	0					MDP generatioed		
11	NBE	147	0					MDP generatioed		
12	HPM	33	0					MDP generatioed		
13	RHP	148	0					No USE		
14	SG	8	0							
					GC	8	1	Gate Poly transistor		
					GR	8	2	Gate POLY resistor		
15	PM	35	0					MDP generatioed		
16	NM	7	0					MDP generatioed		
17	R	12	0					MDP generatioed		
18	PSD	9	0					MDP generatioed		
19	NSD	28	0					MDP generatioed		
20	CONT	11	0		CO	11	0			
21	M1	13	0		M1	13	0			
22	TC	19	0		V1	19	0			
23	M2	20	0		M2	20	0			
24	PRO	14	0		PO	14	0			
	TXM1	48	0		TXM1	48	0	M1 Label		
	PIN	48	1		PIN	48	1	M1 Pin		
	TXM2	49	0		TXM2	49	0	M2 Label		
	MASK	63	0		MASK	63	0	Waiver region		
	PTECT	63	1		PTECT	63	1	Protect region (LOGO)		
	SCRB	80	0		SCRB	80	0	Waiver region (EDGE SEAL)		
	prBoundary	235	0		prBoundary	235	0	Cell boundary		

Cat-1: Prohibited



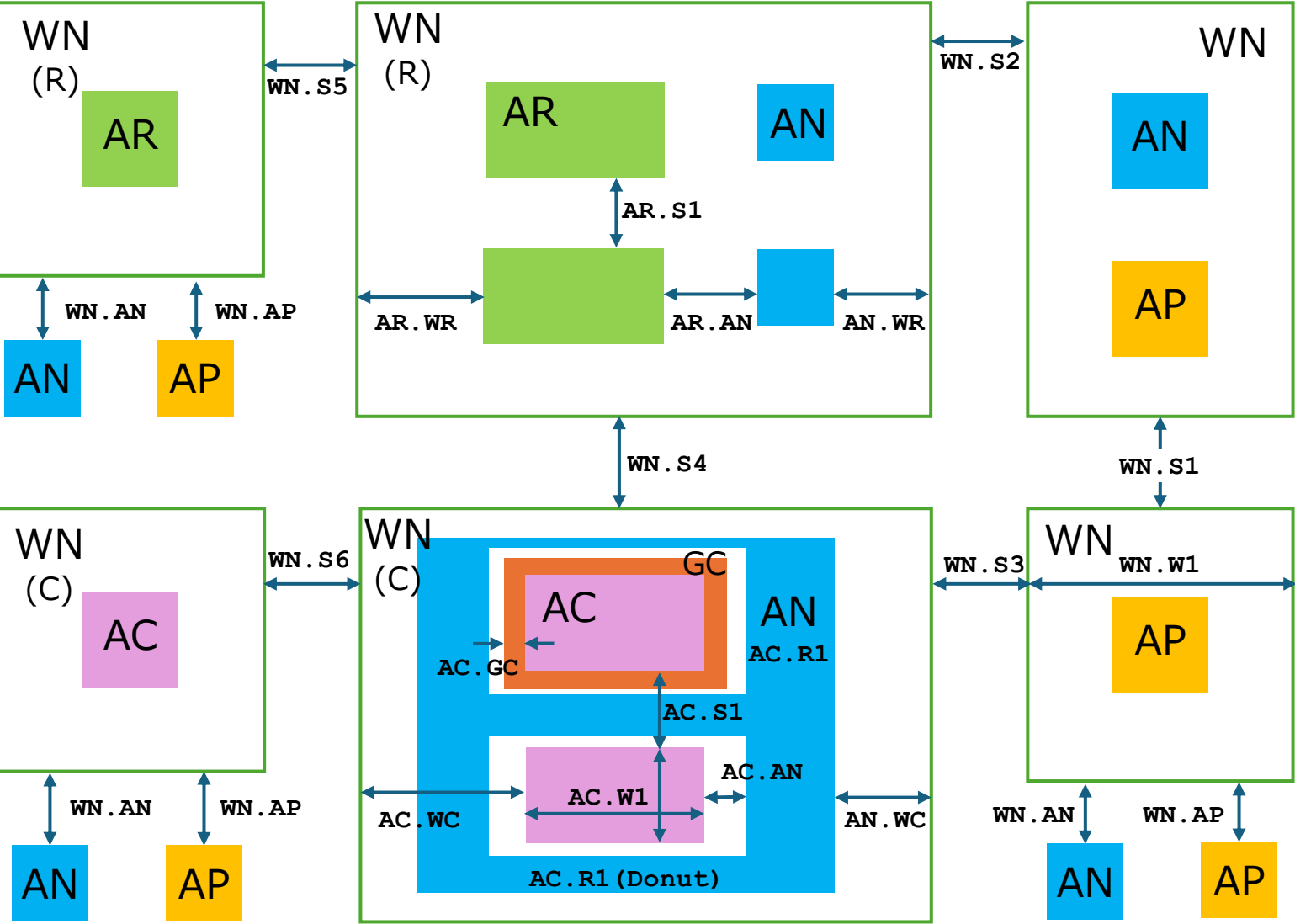
Rule	L1	L2	Func	MIN	MAX
WN.Z1	WN (R)	AP	Prohibit		
WN.Z2	WN (C)	AP	Prohibit		
WN.Z3	WN (C)	AR	Prohibit		
CO.Z1	CO	AA+GC+GR	Require		
CO.Z3	CO	M1	Require		
V1.Z1	V1	M1	Require		
V1.Z2	V1	M2	Require		
PO.Z1	PO	M1	Require		
PO.Z2	PO	M2	Require		
PO.Z3	PO	V1 (P)	Contain		

Function:

Wmin/max: L1 Width min/max
Smin/max: L1 Space min/max (L1=L2)
Smin/max: L1 Separate L2 min/max
Emin/max: L1 Enclosed L2 min/max
Fmin/max: L1 Enclosing L2 min/max
Wfix: L1 Width equal
Sfix: Space equal
Efix: Fixed enclosure
Donut: L1 Souuounded by L2
Prohibit: Must not overlap
Require: Must not be outside
Contain: Must contain
Rect: Must be a rectangle

AA : AP+AN+AC+AR
WN(R) : WN covering AR
WN(C) : WN covering AC
V1(P) : V1 interacting PO

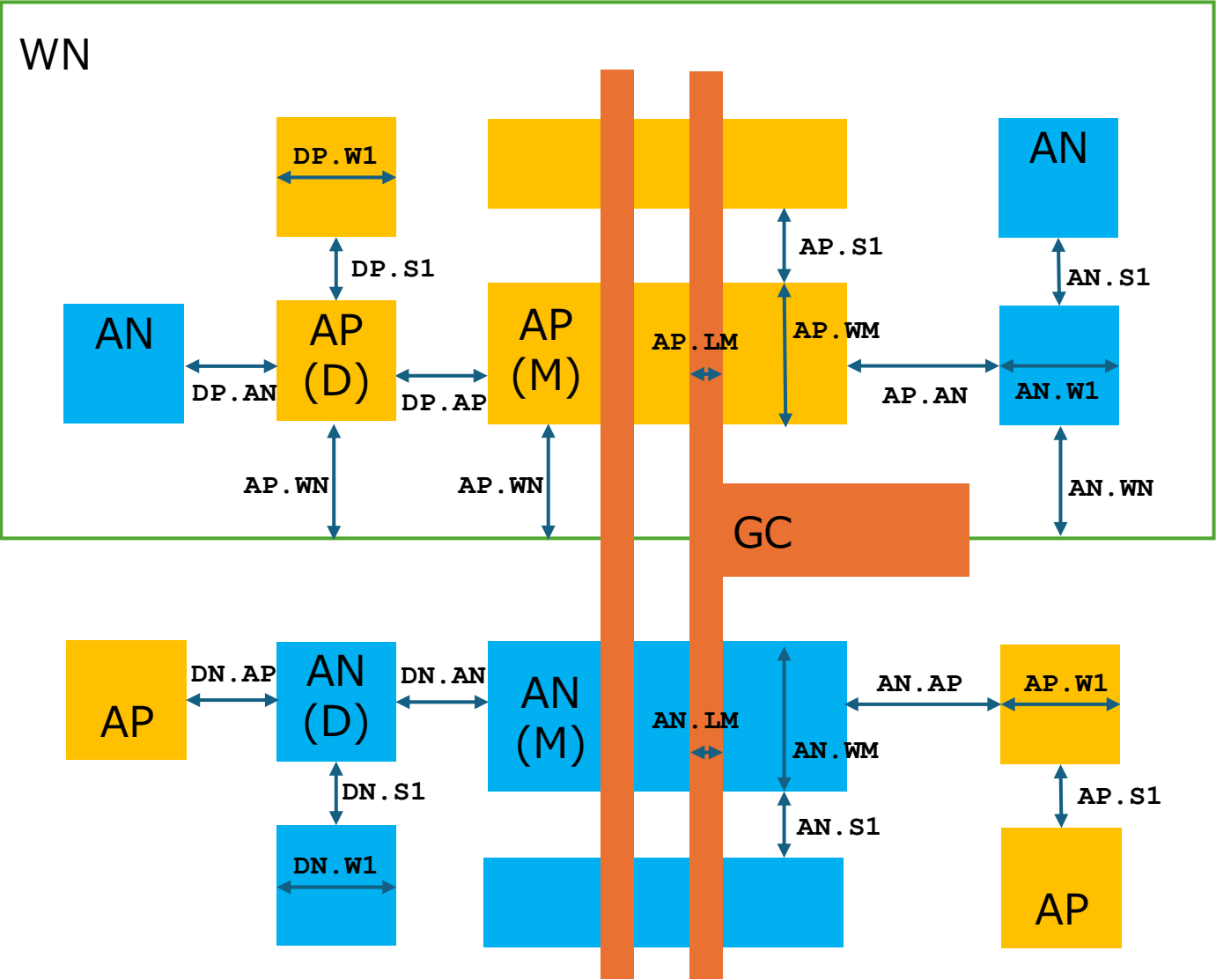
Cat-2: WN to AP/AN/AR/AC related



Rule	L1	L2	Func	MIN	MAX
WN.W1	WN		Wmin	8.0	
WN.S1	WN	WN	Smin	12.0	
WN.S2	WN (R)	WN	Smin	9.5	
WN.S3	WN (C)	WN	Smin	12.0	
WN.S4	WN (R)	WN (C)	Smin	9.5	
WN.S5	WN (R)	WN (R)	Smin	8.0	
WN.S6	WN (C)	WN (C)	Smin	12.0	
WN.AP	WN	AP	Smin	5.0	
WN.AN	WN	AN	Smin	10.0	
AR.S1	AR	AR	Smin	4.0	
AR.WR	AR	WN (R)	Emin	10.0	
AN.WR	AN	WN (R)	Emin	5.0	
AR.AN	AR	AN	Smin	4.0	
AC.W1	AC		Rect	28.5	120.0
AC.S1	AC	AC	Smin	6.4	
AC.WC	AC	WN (C)	Emin	8.0	
AN.WC	AN	WN (C)	Emin	3.0	
AC.AN	AC	AN	Sfix	2.8	2.8
AC.GC	AC	GC	Efix	1.4	1.4
AC.R1	AC	AN (C)	Donut	0.0	

WN(R) : WN covering AR
WN(C) : WN covering AC
AN(C) : AN in WN(C)

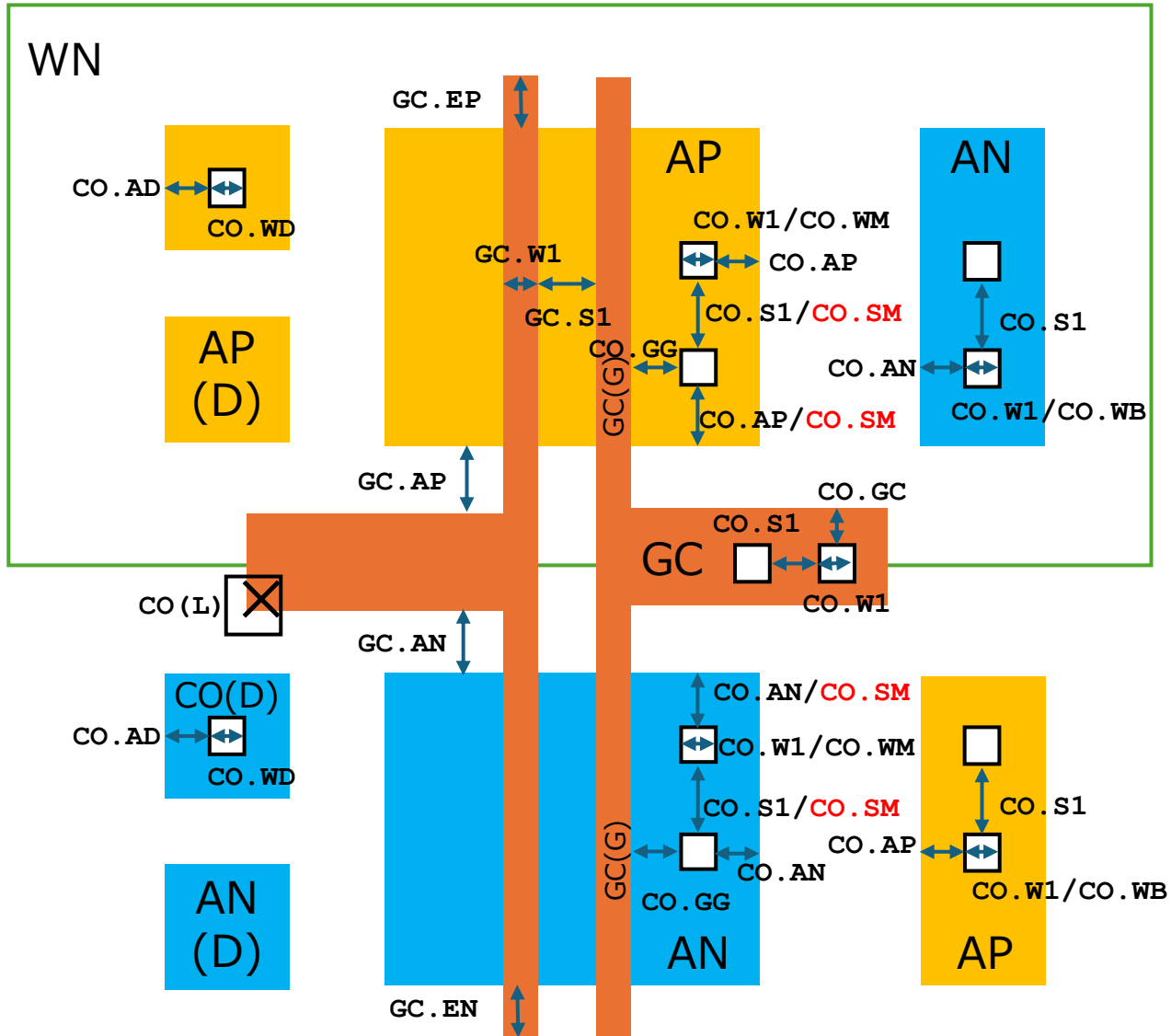
Cat-3: MOS transistor related



Rule	L1	L2	Func	MIN	MAX
AP.W1	AP		Wmin	1.4	
AP.S1	AP	AP	Smin	1.4	
AP.AN	AP	AN	Smin	2.8	
AP.WM	PMOS	AP-GC	Wmin/max	3.4	60.0
AP.LM	PMOS	GC-AP	Lmin/max	1.0	30.0
AP.WN	AP	WN	Emin	7.0	
DP.W1	DP		Wmin	2.8	
DP.S1	DP	DP	Smin	1.4	
DP.AP	DP	AP (M)	Smin	2.8	
DP.AN	DP	AN	Smin	2.8	
AN.W1	AN		Wmin	1.4	
AN.S1	AN	AN	Smin	1.4	
AN.AP	AN	AP	Smin	2.8	
AN.WM	NMOS	AN-GC	Wmin/max	3.4	60.0
AN.LM	NMOS	GC-AN	Lmin/max	1.0	30.0
AN.WN	AN	WN (M)	Emin	5.0	
DN.W1	DN		Wmin	2.8	
DN.S1	DN	DN	Smin	1.4	
DN.AP	DN	AP	Smin	2.8	
DN.AN	DN	AN (M)	Smin	2.8	

PMOS : AP&GC-ESD
NMOS : AN&GC-ESD
DP/AP (D) : AP not interacting(GC) inside WN
DN/AN (D) : AN not interacting(GC) outside WN
AN (M) : AN-AN (D)
AP (M) : AP-AP (D)
WN (M) : WN- (WN covering AC)

Cat-4: Contact and Poly related



Rule	L1	L2	Func	MIN	MAX
GC.W1	GC		Wmin	1.0	
GC.S1	GC	GC	Smin	1.2	
GC.AP	GC	AP	Smin	0.4	
GC.AN	GC	AN	Smin	0.4	
GC.EP	AP	GC	Emin	1.2	
GC.EN	AN	GC	Emin	1.2	
CO.W1	CO		Rect	1.0	
CO.WM	CO (M)		Wfix	1.0	1.0
CO.WB	CO (B)		Wfix	1.0	1.0
CO.S1	CO	CO	Smin	1.0	
CO.SM	CO (M)	CO (M)	Sfix	1.0	1.0
CO.WD	CO (D)		Wfix	1.2	1.2
CO.SD	CO (D)	CO	Smin	1.2	
CO.AD	CO (D)	AP+AN	Emin	1.2	
CO.AP	CO	AP	Emin	0.8	
CO.AN	CO	AN	Emin	0.8	
CL.GC	CO (L)	GC+GR	Prohibit		
CO.GG	CO (M)	GC	Sfix	1.0	1.0
CO.GC	CO	GC+GR	Emin	0.8	

PMOS : AP&GC-ESD

NMOS : AN&GC-ESD

AP (D) : AP not interacting GC

AN (D) : AN not interacting GC

CO (M) : CO on PMOS/NMOS

$$\text{CO}(B) : \text{CO on BG}$$
$$\text{CO}(D) : \text{CO on AP}(D) / \text{AN}(D)$$

CO (L) : CO (Area $\geq 1.5\mu\text{m}^2$)

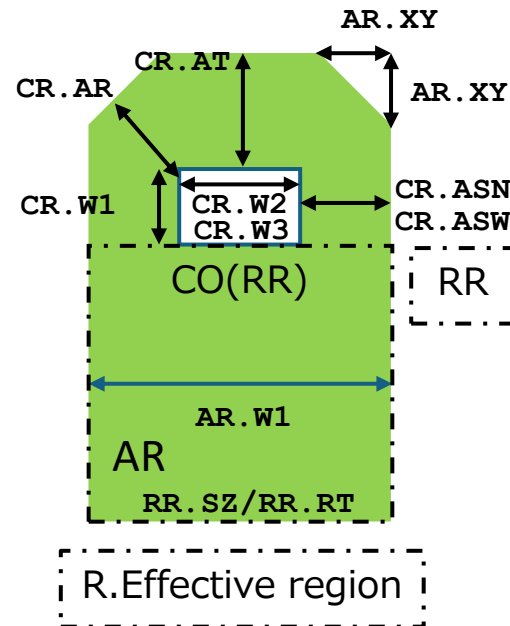
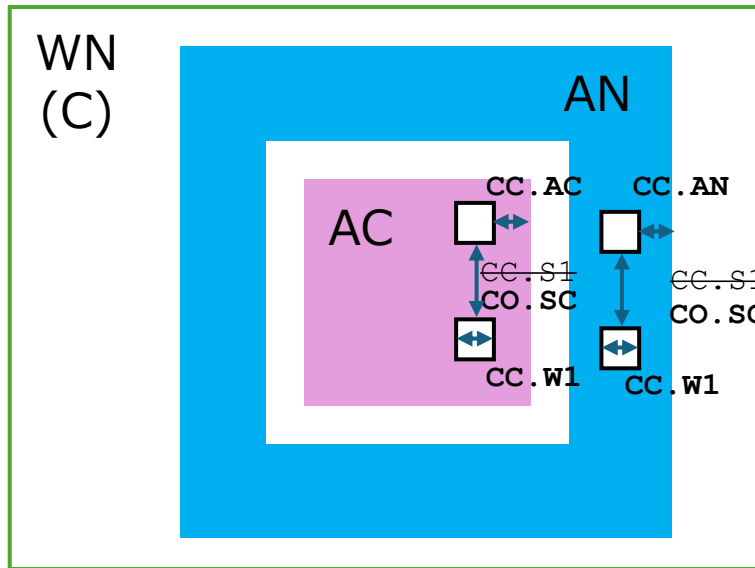
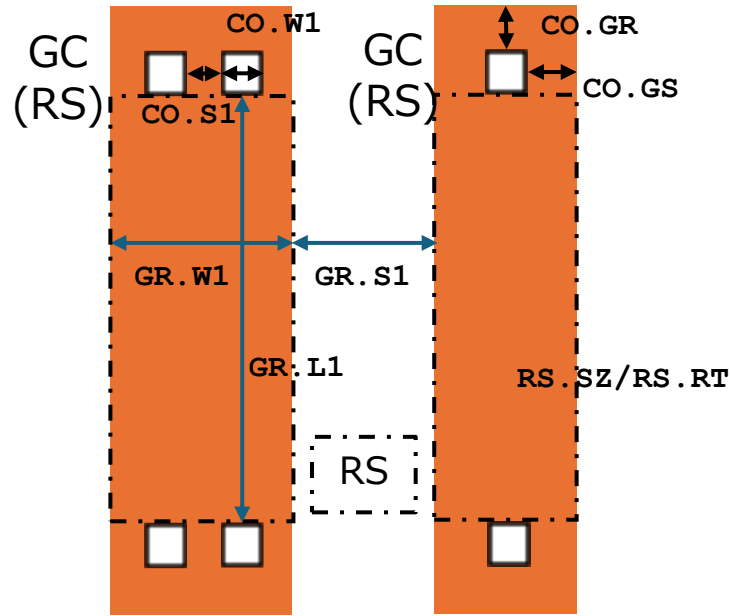
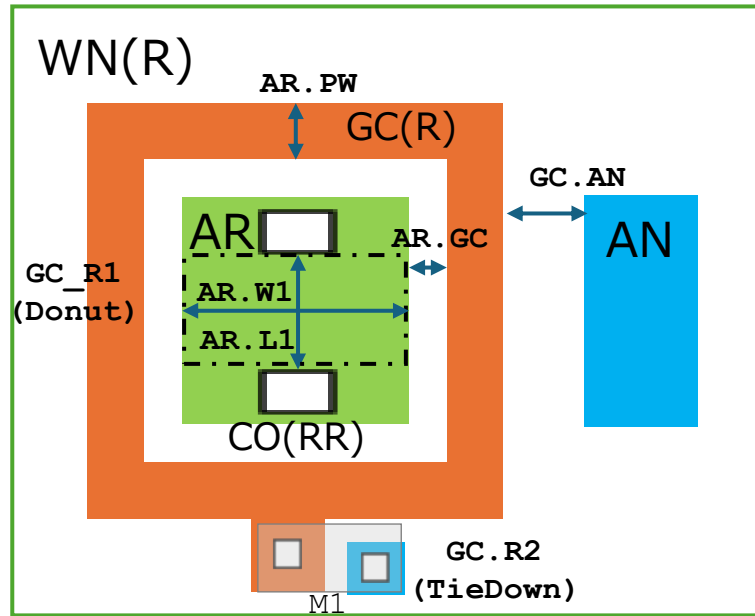
GP : AN&WN

GN : AP-WN

WC : WN covering AC

$$BG : (GP+GN) - WC$$

Cat-5: Resistors(AR/GR) and Capacitor(AC) related

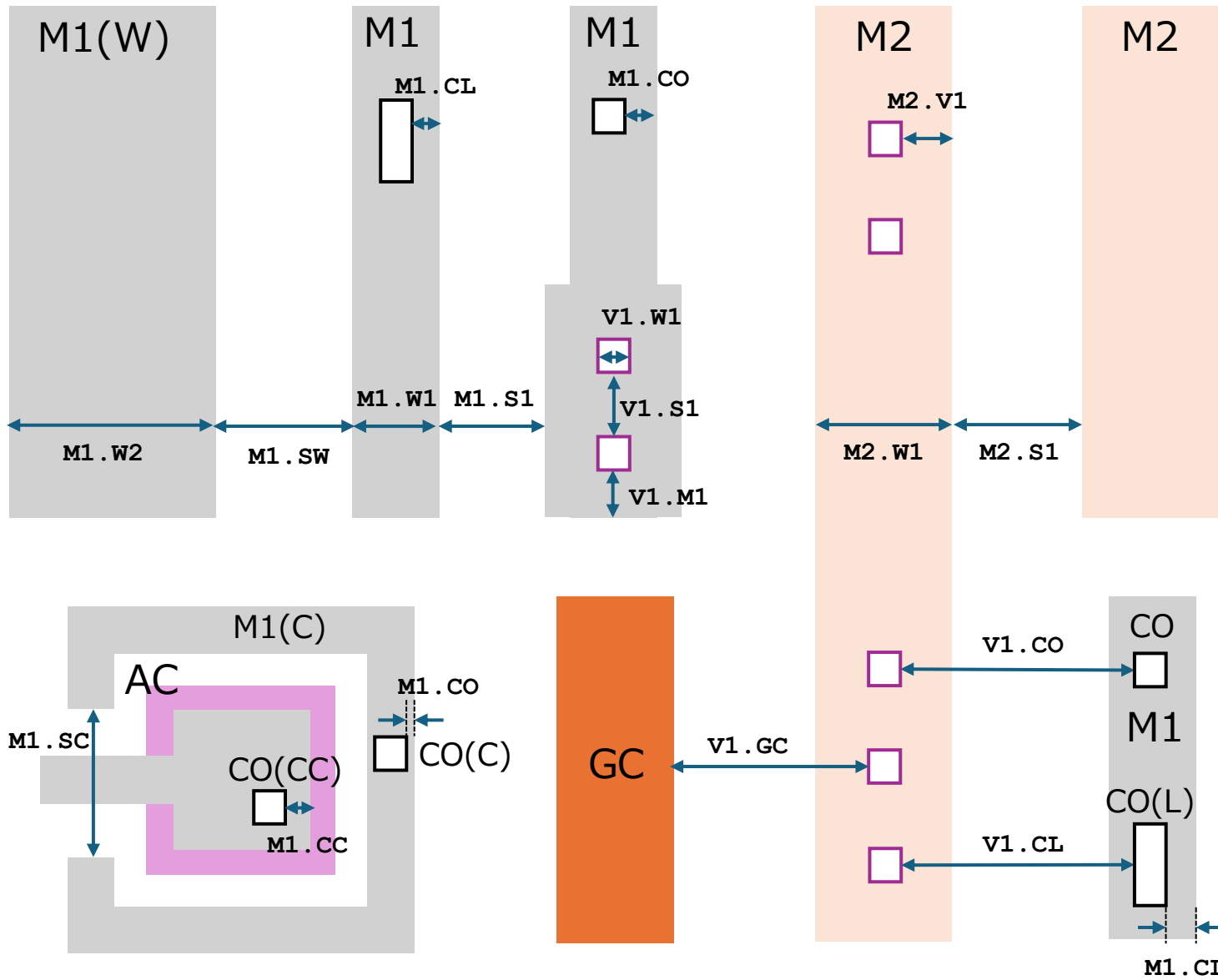


Rule	L1	L2	Func	MIN	MAX
AR.W1	RR	RR (W)	Wmin/max	2.8	20.0
AR.L1	RR	RR (L)	Lmin/max	13.0	100.0
RR.SZ	RR	RROK	Contain		
RR.RT	RR		Rect		
AR.GC	GC	AR	Smin	1.0	
AR.PW	GC (R)		Wmin	2.0	
AR.XY	AR	Bevel	XYmin	1.0	
GC.AN	GC (R)	AN	Smin	???	???
GC.R1	AR	GC (R)	Donut	0.0	
GC.R2	GC (R)	WN (R)	TieDown	0.0	
GR.W1	RS	RS (W)	Wmin/max	4.0	20.0
GR.L1	RS	RS (L)	Lmin/max	20.0	100.0
RS.SZ	RS	RSOK	Require		
RS.RT	RS		Rect		
GR.S1	RS	RS	Smin	2.0	
CR.W1	CO (RR)		Wmin	1.0	
CR.W2	CO (RRW)		Lmin/max	1.6	17.6
CR.W3	CO (RRN)		Wfix	1.0	1.0
CR.AR	CO (RR)	AR	Emin	0.8	
CR.AT	CO (RR)	AR (T)	Emin	1.5	
CR.ASN	CO (RRN)	ARN (S)	Efix	0.9	
CR.ASW	CO (RRW)	ARW (S)	Efix	1.2	
CC.W1	CO (C)		Wfix	1.2	1.2
CC.S1	CO (C)	CO (C)	SFIX	1.2	1.2
CO.SC	CO (C)	CO	Smin	1.2	
CC.AC	CO (C)	AC	Emin	2.9	
CC.AN	CO (C)	AN	Emin	1.2	
ES.S1	CO (RS)	CO (RS)	SFIX	1.0	1.0
CO.GR	CO	GR	Emin	0.8	
CO.GS	CO (RSM)	GR (S)	Efix	1.5	1.5
CO.RT	CO (RSM)		Rect		

RR(W) : Width of AR effective
 RR(L) : Length of AR effective
 RROK : Allowed W=2.8,4,6,12,20
 GC(R) : GC in WN(R)
 RS(W) : Width of GR effective
 RS(L) : Length of GR effective
 RSOK : Allowed W=4,6,12,20
 WC : WN interacting AC
 AR(T) : AR top fringe
 CO(RR) : CO on AR
 CO(RRW) : CO on AR (W>=4.0)
 CO(RRN) : CO on AR (W=2.8)
 ARN(S) : AR side fringe (W=2.8)
 ARW(S) : AR side fringe (W>=4.0)
 CO(C) : CO on WC
 CO(RS) : CO on GR
 GR(S) : GR side fringe
 CO(RSM) : CO(sized) & GR(S)

*Strike-through rules are not yet implemented

Cat-6: M1/V1/M2 related

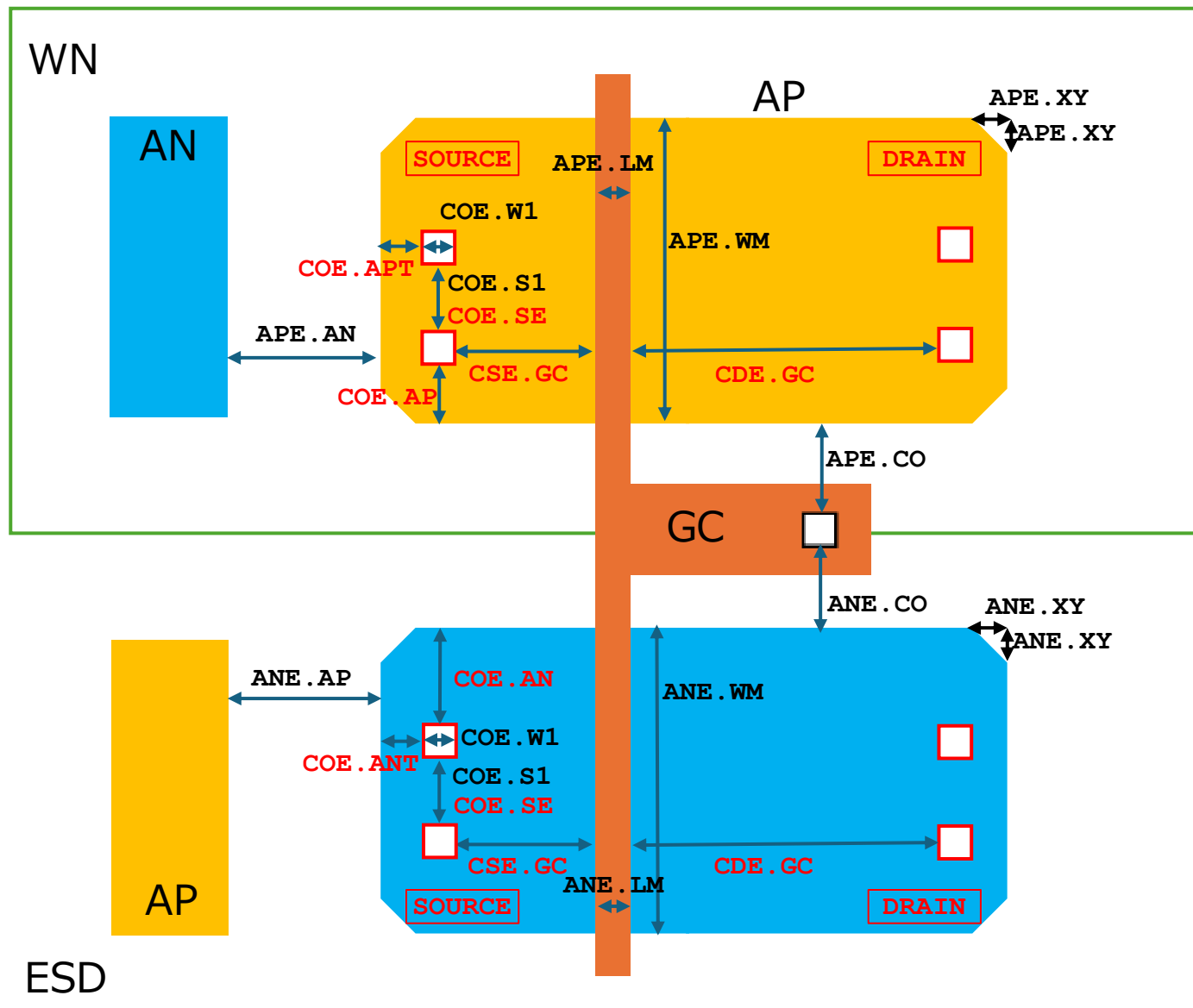


Rule	L1	L2	Func	MIN	MAX
M1.W1	M1		Wmin	1.8	
M1.W2	M1 (W)		Wmin	10.0	
M1.S1	M1	M1	Smin	1.4	
M1.SC	M1 (C)	M1 (C)	Smin/Smax	10.0	10.0
M1.CO	M1	CO	Fmin	0.8	
M1.CC	M1	CO (CC)	Fmin	1.3	
M1.CL	M1	CO (L)	Fmin	1.2	
M1.SW	M1 (W)	M1	Smin	2.0	
V1.W1	V1		Wfix	1.4	1.4
V1.S1	V1	V1	Smin	1.5	
V1.M1	V1	M1	Emin	1.0	
V1.GC	V1	GC	Smin	1.2	
V1.CO	V1	CO	Smin	1.0	
V1.CL	V1	CO (L)	Smin	1.4	
M2.W1	M2		Wmin	3.0	
M2.S1	M2	M2	Smin	2.0	
M2.V1	M2	V1	Fmin	1.0	

M1 (W) : Wide M1 (W>=10.0)
WN (C) : WN covering AC
M1 (C) : M1 interacting (AN & WN (C))
CO (C) : CO on M1 (C) (W=1.2um)
CO (CC) : CO on AC (W=1.2um)
CO (L) : CO large contact (Area>=1.5um²)

*Strike-through rules are not yet implemented

Cat-7: ESD MOS transistor related

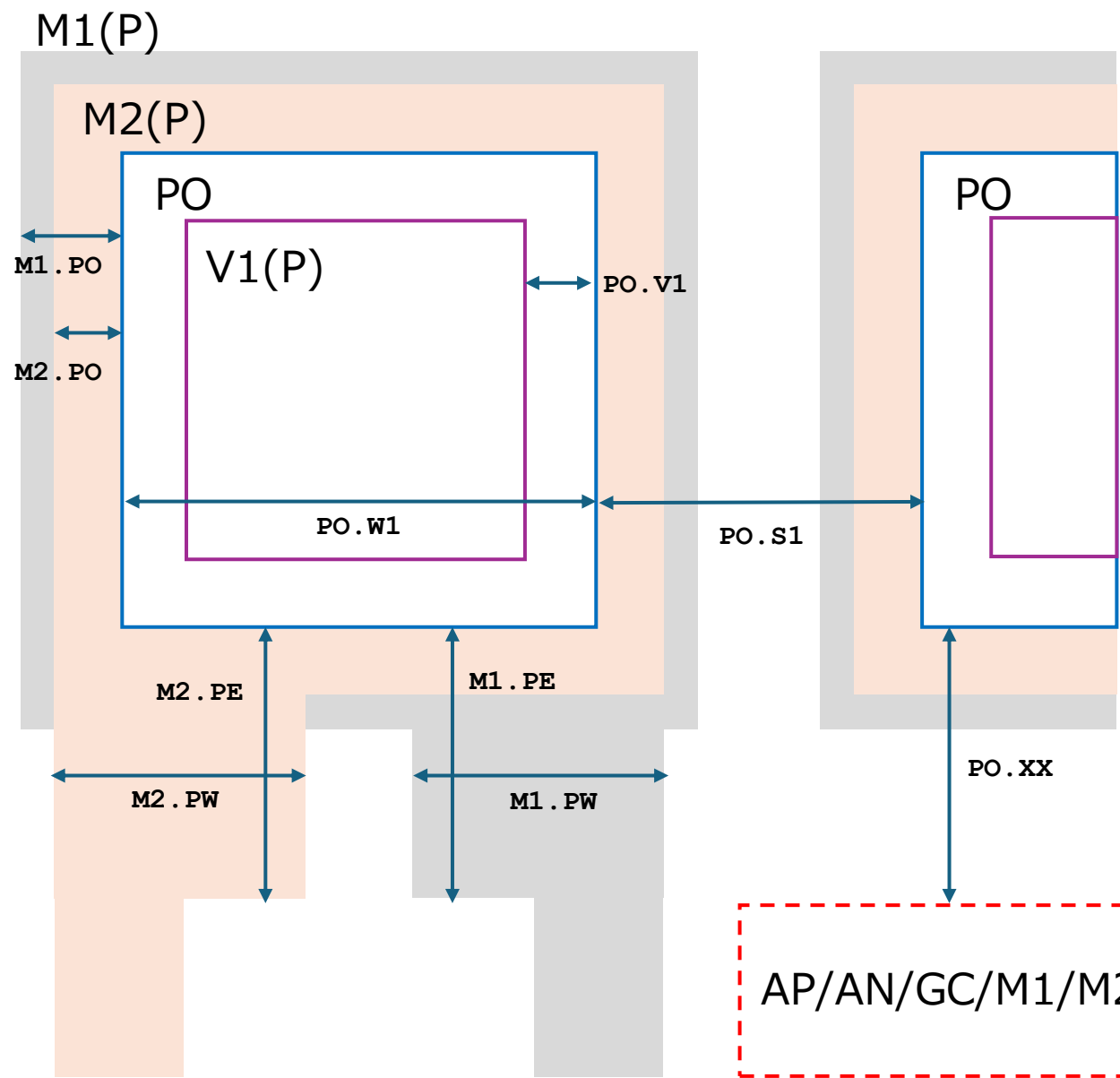


Rule	L1	L2	Func	MIN	MAX
APE.WM	PMOSE	APE-GC	Wmin/max	11.0	60.0
APE.LM	PMOSE	GC-APE	Lmin/max	2.0	30.0
APE.AN	APE	AN	Smin	10.0	
APE.XY	APE	Bevel	XYmin	1.4	
APE.CO	APE	CO	Smin	2.0	
ANE.WM	NMOSE	ANE-GC	Wmin/max	11.0	60.0
ANE.LM	NMOSE	GC-ANE	Lmin/max	2.0	30.0
ANE.AP	ANE	AP	Smin	10.0	
ANE.XY	ANE	Bevel	XYmin	1.4	
ANE.CO	ANE	CO	Smin	2.0	
COE.W1	CO (E)		Wfix	3.0	3.0
COE.S1	CO (E)	CO	Smin	1.6	
COE.SE	CO (E)	CO (E)	Sfix	1.6	1.6
CDE.GC	CD (E)	GC	Sfix	7.0	
CSE.GC	CS (E)	GC	Sfix	3.0	
COE.AP	CO (E)	AP	Emin	2.5	
COE.AN	CO (E)	AN	Emin	2.5	
COE.APT	CO (E)	AP (T)	Emin	4.0	
COE.ANT	CO (E)	AN (T)	Emin	4.0	

MPE	: AP interacting(GC)	& WN	& ESD
MNE	: AN interacting(GC)	- WN	& ESD
PMOSE	: MPE	& GC	
NMOSE	: MNE	& GC	
APE	: MPE		
ANE	: MNE		CO (E) : C
AP(T)	: AP Side Edges		CD (E) : C
AN(T)	: AN Side Edges		CS (E) : C

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CO(E) : CO on ESD
CD(E) : CO on ESD/Drain
CS(E) : CO on ESD/Source
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Cat-8: PAD related



Rule	L1	L2	Func	MIN	MAX
PO.W1	PO		Wmin	70.0	
PO.S1	PO	PO	Smin	64.0	
M2.PO	M2 (P)	PO	Fmin	5.0	
M1.PO	M1 (P)	PO	Fmin	5.0	
PO.V1	PO	V1 (P)	Emin	10.0	
PO.AP	PO	AP	Smin	14.0	
PO.AN	PO	AN	Smin	14.0	
PO.GC	PO	GC	Smin	14.0	
PO.M1	PO	M1	Smin	14.0	
PO.M2	PO	M2	Smin	14.0	
M2.PE	M2		EXTL	19.0	
M2.PW	M2		EXTW	40.0	
M1.PE	M2		EXTL	19.0	
M1.PW	M2		EXTW	40.0	

** Not yet implemented

M1 (P) : M1 covering PO
M2 (P) : M2 covering PO
V1 (P) : V1 through M1 (P) -M2 (P)

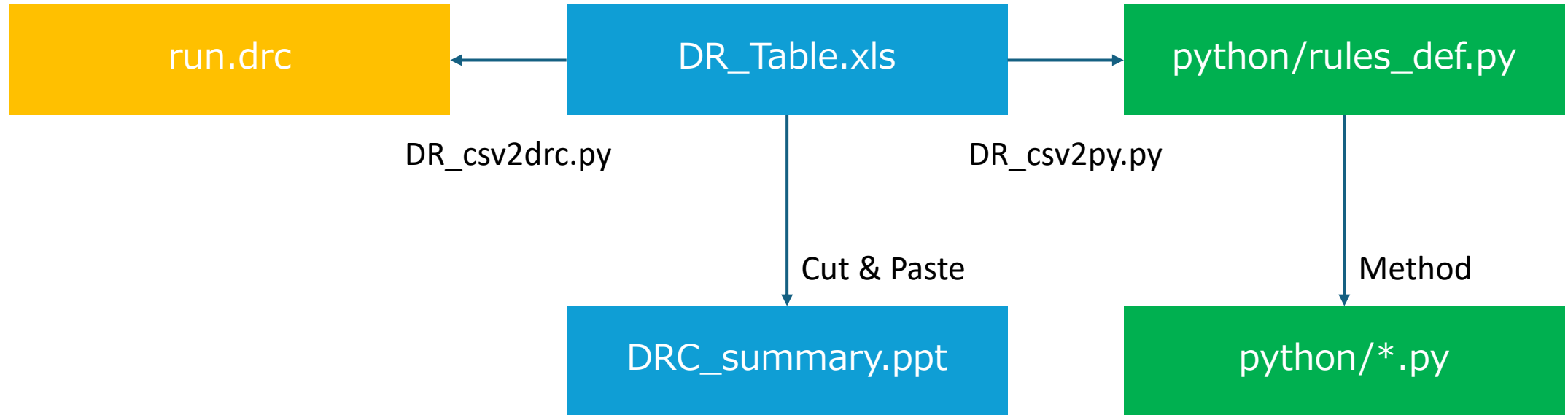
Design Rule control

DRC integrity

DRC

Document

PCell



ALL Critical Dimension parameters are referenced by 'XX.YY' rule's name.

Design to Tapeout Process

Mask Data Preparation

Designer

local



[push]

remote



[action]
Precheck
DRC
MDP



[Import artifact]

OpenSUSI

[clone]



TR-1um_MPW_XXX

[action]
Logo insertion
Aggregation
DRC



Tokai Rika

Signoff DRC

Fabrication

PCM TEST
Wafer out

Download GDSII

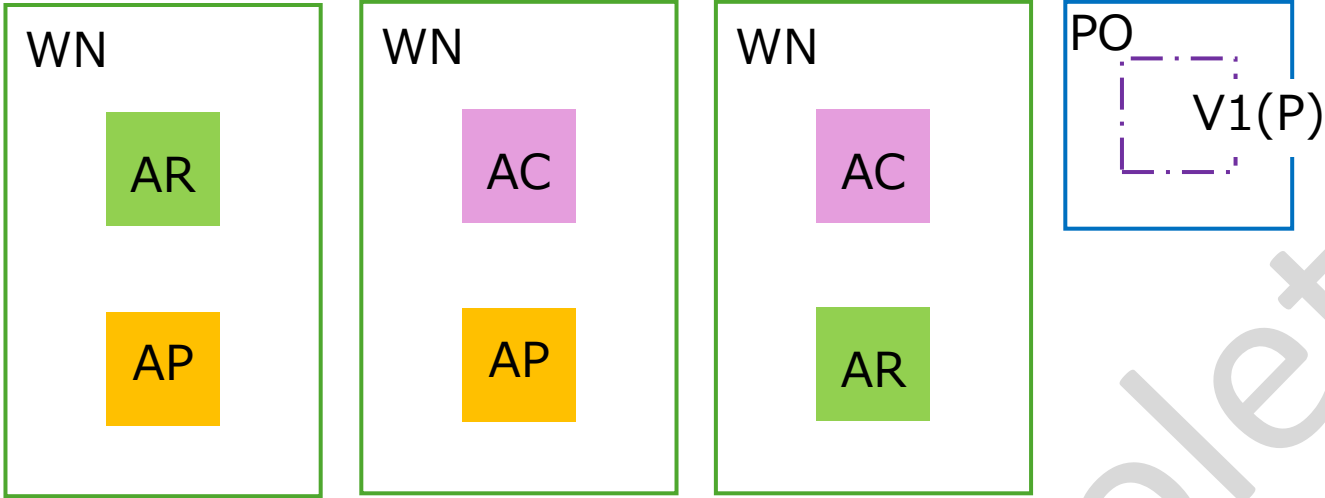
Obsolete slides

2026/06/30

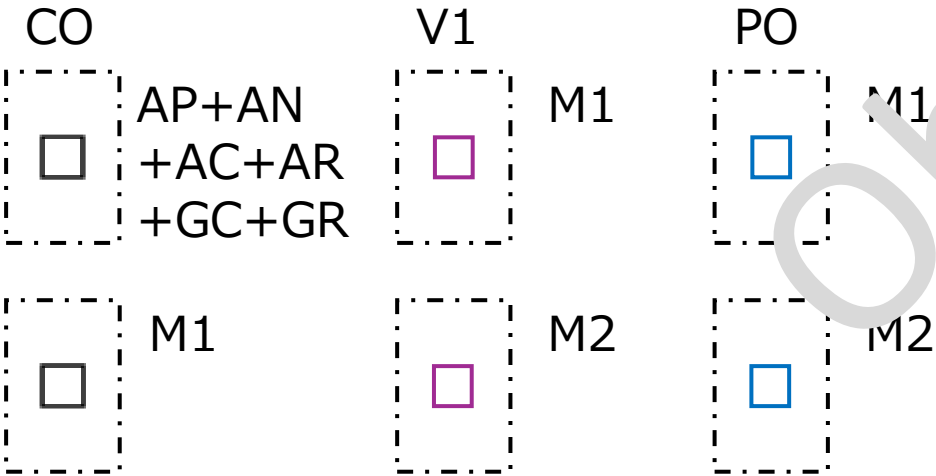
Reflected GitHub issues

<https://github.com/OpenSUSI/TR-1um/issues>

Cat-1: Prohibited



Rule	L1	L2	Func	MIN	MAX
WN.Z1	WN (R)	AP	Prohibit		
WN.Z2	WN (C)	AP	Prohibit		
WN.Z3	WN (C)	AR	Prohibit		
CO.Z1	CO	AA+GC+GR	Require		
CO.Z3	CO	M1	Require		
V1.Z1	V1	M1	Require		
V1.Z2	V1	M2	Require		
PO.Z1	PO	M1	Require		
PO.Z2	PO	M2	Require		
PO.Z3	PO	V1 (P)	Contain		

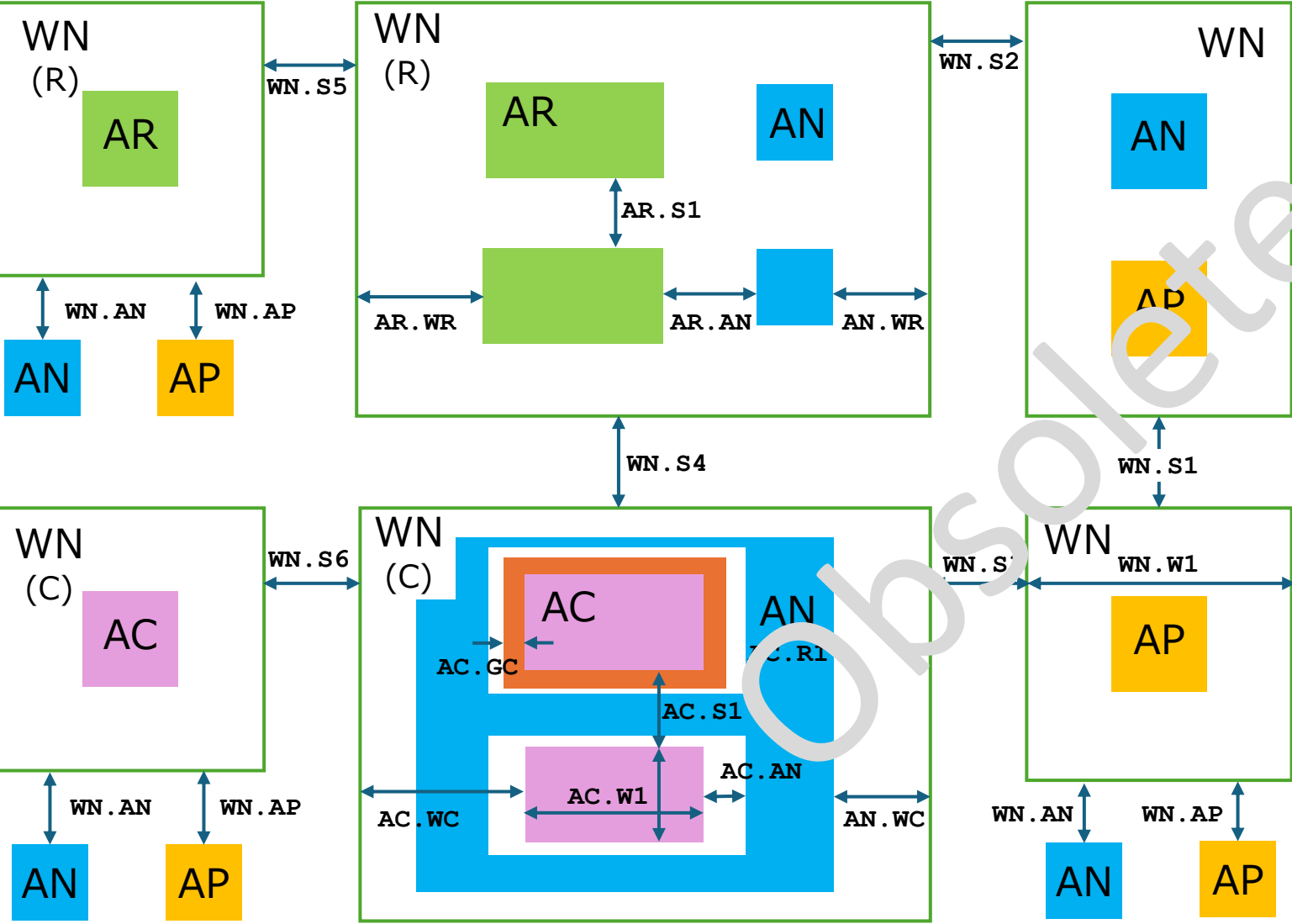


Function:

Wmin/max: L1 Width min/max
Smin/max: L1 Space min/max (L1=L2)
Smin/max: L1 Separate L2 min/max
Emin/max: L1 Enclosed L2 min/max
Fmin/max: L1 Enclosing L2 min/max
ECmin: MOS Endcap min
Wfix: L1 Width equal
Sfix: Space equal
Efix: Fixed enclosure
Donut: L1 Surrounding by L2
Prohibit: Must not overlap
Require: Must not be outside
Contain: Must contain
Rect: Must be a rectangle

WN(R) : WN covering AR
WN(C) : WN covering AC
V1(P) : V1 interacting PO

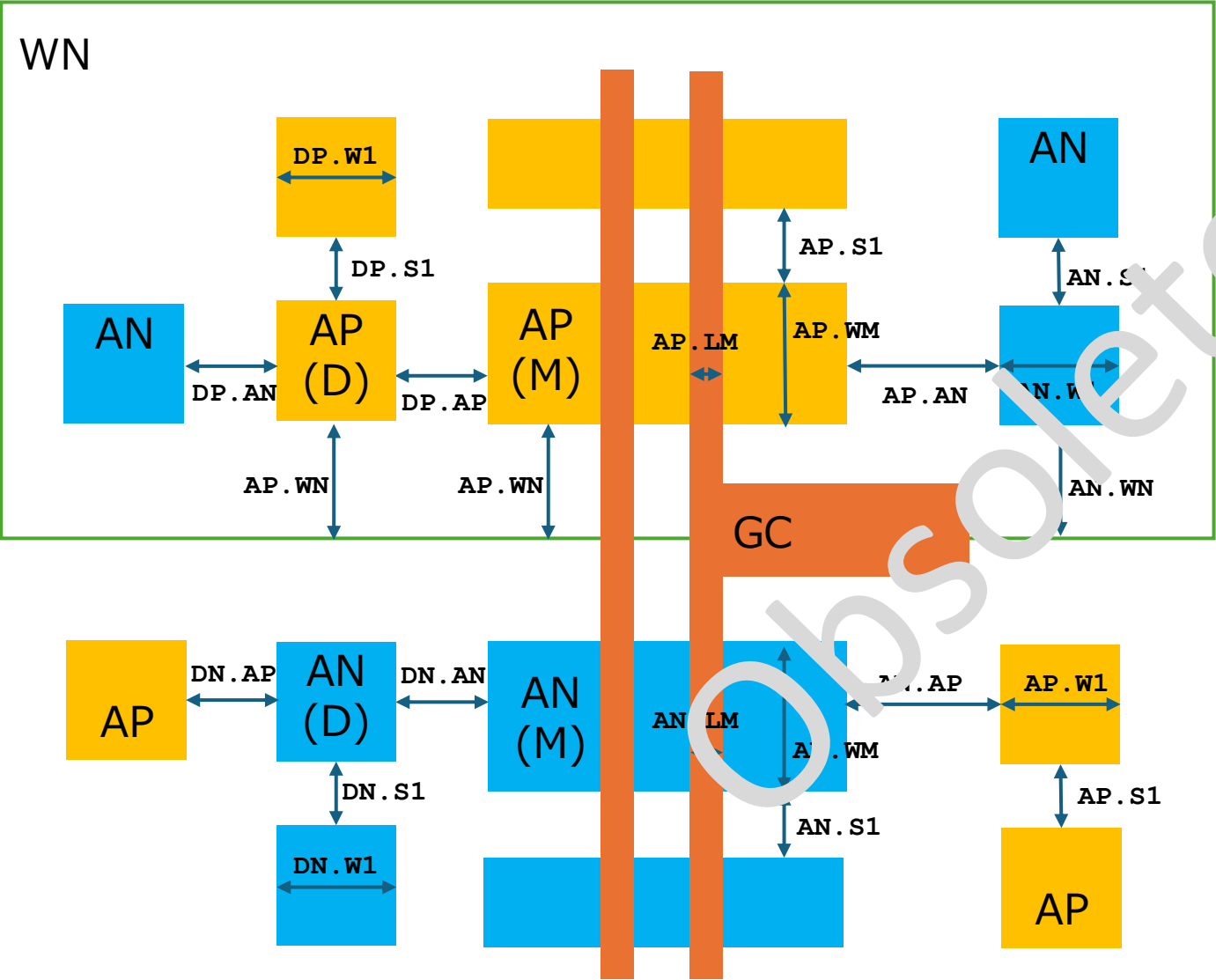
Cat-2: WN to AP/AN/AR/AC related



Rule	L1	L2	Func	MIN	MAX
WN.W1	WN		Wmin	8.0	
WN.S1	WN	WN	Smin	12.0	
WN.S2	WN (R)	WN	Smin	9.5	
WN.S3	WN (C)	WN	Smin	12.0	
WN.S4	WN (R)	WN (C)	Smin	9.5	
WN.S5	WN (R)	WN (R)	Smin	8.0	
WN.S6	WN (C)	WN (C)	Smin	12.0	
WN.AP	WN	AP	Smin	5.0	
WN.AN	WN	AN	Smin	10.0	
AR.S1	AR	AR	Smin	4.0	
AR.WR	AR	WN (R)	Emin	10.0	
AN.WR	AN	WN (R)	Emin	5.0	
AR.AN	AR	AN	Smin	4.0	
AC.W1	AC		Rect	28.5	120.0
AC.S1	AC	AC	Smin	6.4	
AC.WC	AC	WN (C)	Emin	8.0	
AN.WC	AN	WN (C)	Emin	3.0	
AC.AN	AC	AN	Sfix	2.8	2.8
AC.GC	AC	GC	Efix	1.4	1.4
AC.R1	AC	AN (C)	Donut	0.0	

WN(R) : WN covering AR
WN(C) : WN covering AC
AN(C) : AN in WN(C)

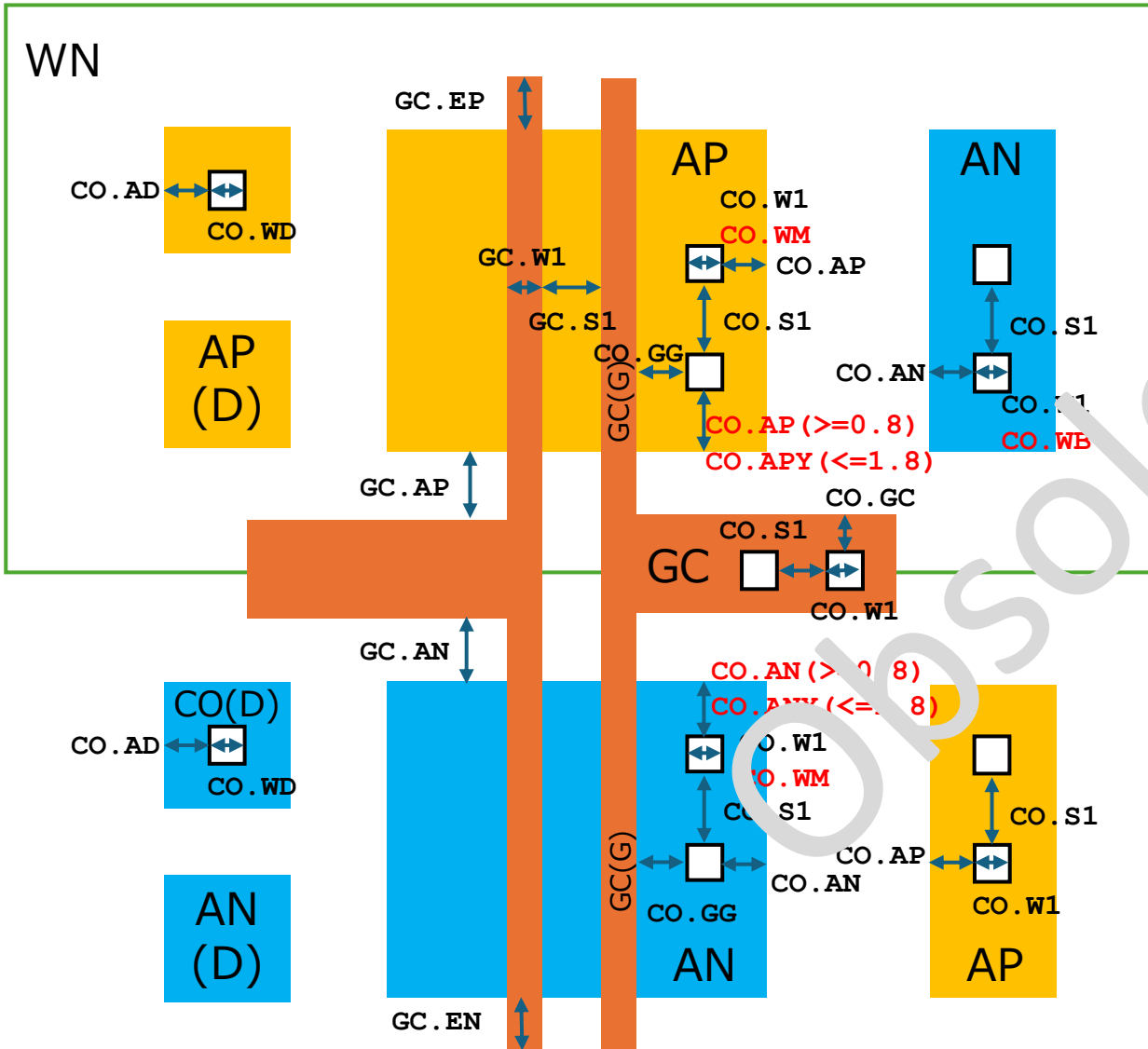
Cat-3: MOS transistor related



Rule	L1	L2	Func	MIN	MAX
AP.W1	AP		Wmin	1.4	
AP.S1	AP	AP	Smin	1.4	
AP.AN	AP	AN	Smin	2.8	
AP.WM	PMOS	AP-GC	Wmin/max	3.4	60.0
AP.LM	PMOS	GC-AN	Lmin/max	1.0	30.0
DP.WN	AP	WN	Emin	7.0	
DP.W1	DP		Wmin	2.8	
DP.S1	DP	DP	Smin	1.4	
DP.AP	DP	AP (M)	Smin	2.8	
DP.AN	DP	AN	Smin	2.8	
AN.W1	AN		Wmin	1.4	
AN.S1	AN	AN	Smin	1.4	
AN.AP	AN	AP	Smin	2.8	
AN.WM	NMOS	AN-GC	Wmin/max	3.4	60.0
AN.LM	NMOS	GC-AN	Lmin/max	1.0	30.0
AN.WN	AN	WN (M)	Emin	5.0	
DN.W1	DN		Wmin	2.8	
DN.S1	DN	DN	Smin	1.4	
DN.AP	DN	AP	Smin	2.8	
DN.AN	DN	AN (M)	Smin	2.8	

PMOS : AP&GC
NMOS : AN&GC
DP/AP(D) : AP not interacting GC inside WN
DN/AN(D) : AN not interacting GC outside WN
AN (M) : AN-AN (D)
AP (M) : AP-AP (D)
WN (M) : WN-(WN covering AC)

Cat-4: Contact and Poly related



Rule	L1	L2	Func	MIN	MAX
GC.W1	GC		Wmin	1.0	
GC.S1	GC	GC	Smin	1.2	
GC.AP	GC	AP	Smin	0.4	
GC.AN	GC	AN	Smin	0.4	
GC.EF	PMOS	GC	ECmin	1.2	
GC.EB	NMOS	GC	ECmin	1.2	
CO.W1	CO		Rect	1.0	
CO.WM	CO (M)		Wfix	1.0	1.0
CO.WB	CO (B)		Wfix	1.0	1.0
CO.S1	CO	CO	Smin	1.0	
CO.WD	CO (D)		Wfix	1.2	1.2
CO.SD	CO (D)	CO	Smin	1.2	
CO.AD	CO (D)	AP+AN	Emin	1.2	
CO.AP	CO	AP	Emin	0.8	
CO.AN	CO	AN	Emin	0.8	
CL.GC	CO (L)	GC+GR	Prohibit		
CO.GG	CO	GC (G)	Smin	1.0	
CO.GC	CO	GC+GR	Emin	0.8	

PMOS : AP&GC

GP : AN&WN

NMOS : AN&GC

GN : AP-WN

AP (D) : AP not interacting GC

WC : WN covering AC

AN (D) : AN not interacting GC

$$\text{BG} : (\text{GP} + \text{GN}) - \text{WC}$$

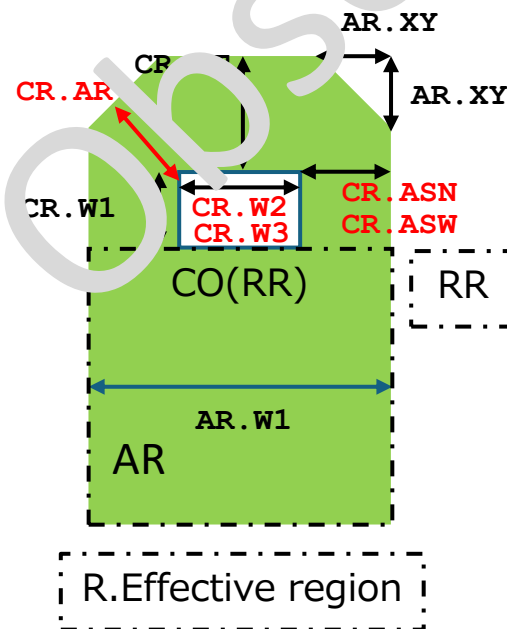
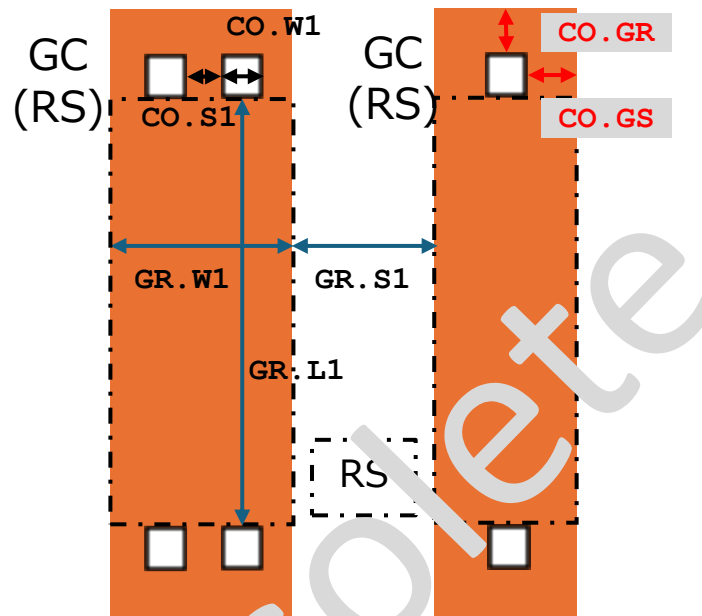
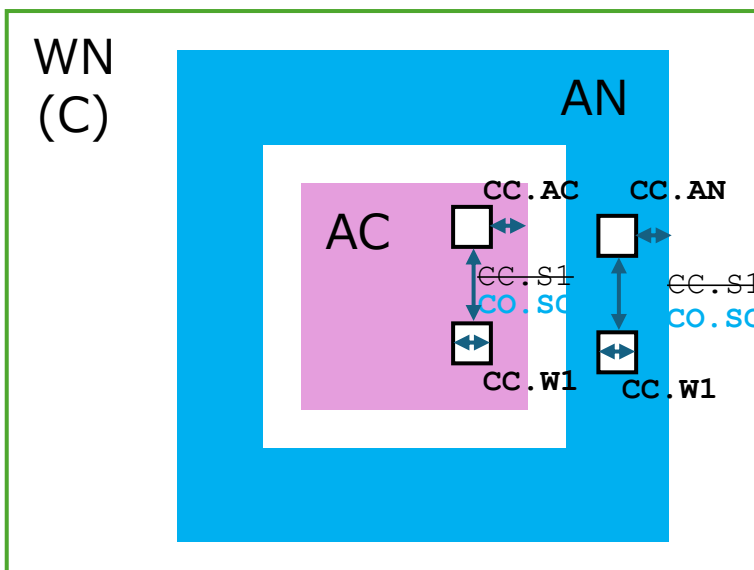
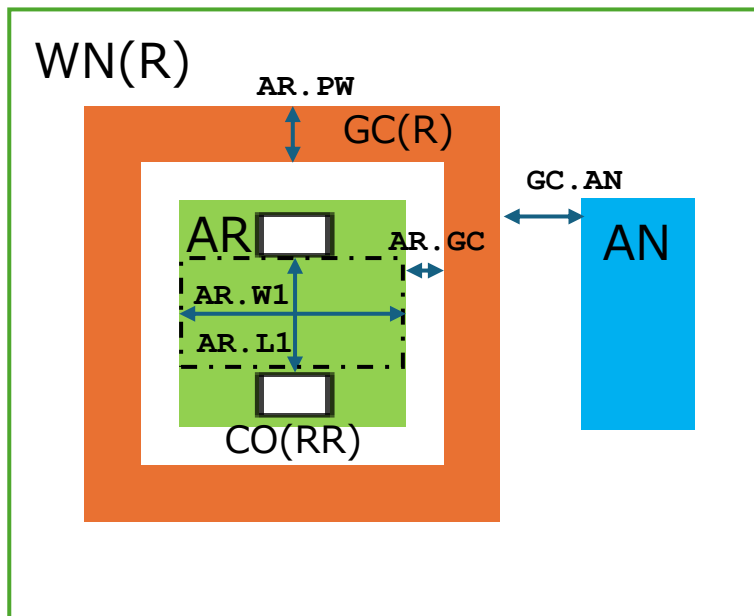
CO (M) : CO on PMOS/NMOS

$$\text{CO}(B) : \text{CO on BG}$$
$$\text{CO}(D) : \text{CO on AP}(D) / \text{AN}(D)$$

CO (L) : CO (Area $\geq 1.5\mu\text{m}^2$)

GC (G) : GC for PMOS/NMOS

Cat-5: Resistors(AR/GR) and Capacitor(AC) related



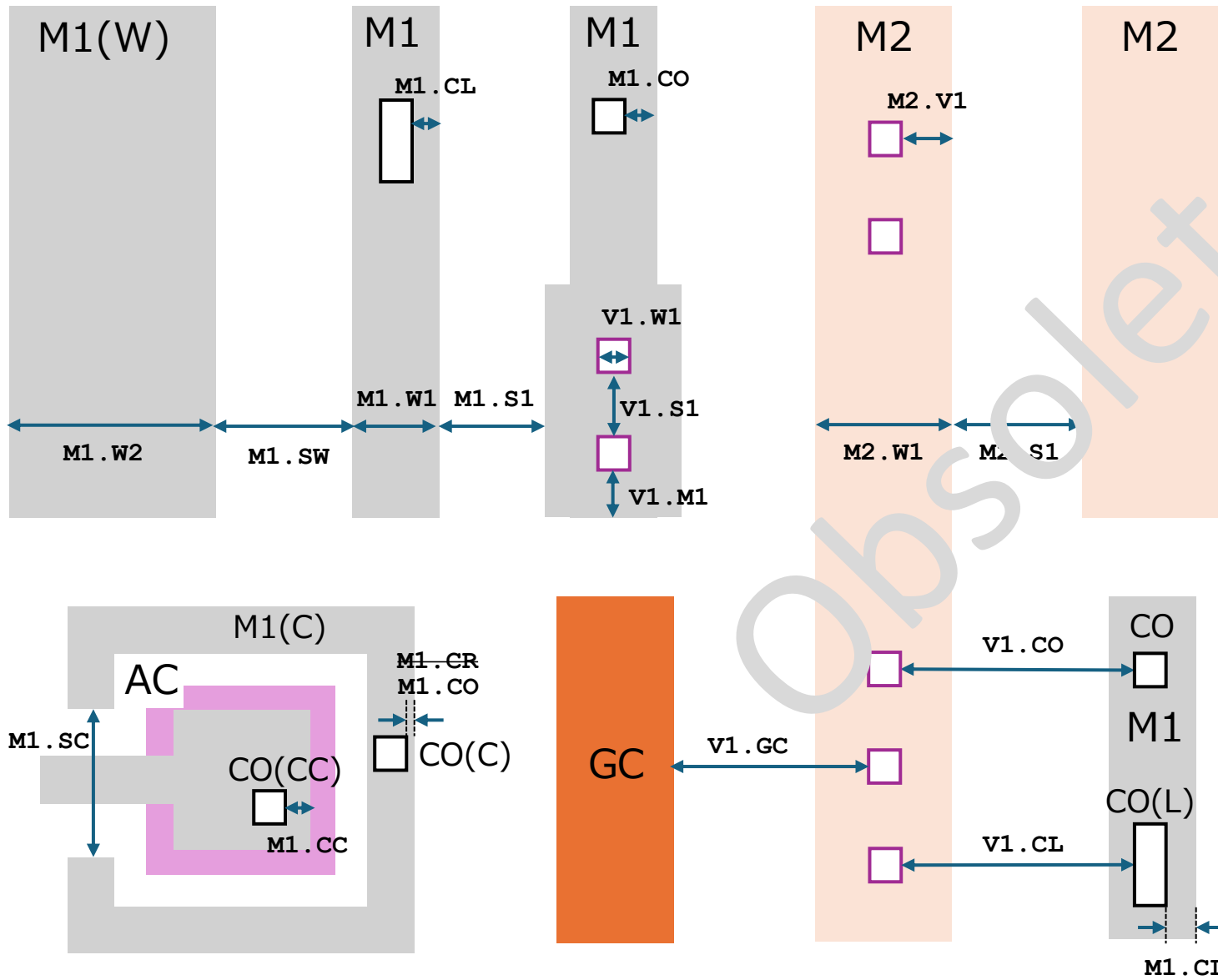
Rule	L1	L2	Func	MIN	MAX
AR.W1	RR	RR (W)	Wmin/max	2.8	20.0
AR.L1	RR	RR (L)	Lmin/max	13.0	100.0
RR.SZ	RR	RROK	Contain		
RR.RT	RR		Rect		
AR.GC	GC	AR	Smin	1.0	
AR.PW	GC (R)		Wmin	2.0	
AR.XY	AR	Bevel	XYmin	1.0	
GC.AN	GC (R)	AN	Smin	???	???
GC.R1	AR	GC (R)	Donut	0.0	
GC.R2	GC (R)	WN (R)	TieDown	0.0	
GR.W1	RS	RS (W)	Wmin/max	4.0	20.0
GR.L1	RS	RS (L)	Lmin/max	20.0	100.0
RS.SZ	RS	RSOK	Require		
RS.RT	RS		Rect		
GR.S1	RS	RS	Smin	2.0	
CR.W1	CO (RR)		Wmin	1.0	
CR.W2	CO (RRW)		Lmin/max	1.6	17.6
CR.W3	CO (RRN)		Wfix	1.0	1.0
CR.AR	CO (RR)	AR	Emin	0.8	
CR.AT	CO (RR)	AR (T)	Emin	1.5	
CR.ASN	CO (RRN)	ARN (S)	Efix	0.9	
CR.ASW	CO (RRW)	ARW (S)	Efix	1.2	
CC.W1	CO (C)		Wfix	1.2	1.2
CC.S1	CO (C)	CO (C)	SFIX	1.2	1.2
CO.SC	CO (C)	CO	Smin	1.2	
CC.AC	CO (C)	AC	Emin	2.9	
CC.AN	CO (C)	AN	Emin	1.2	
ES.S1	CO (RS)	CO (RS)	SFIX	1.0	1.0
CO.GR	CO	GR	Emin	0.8	
CO.GS	CO (RSM)	GR (S)	Efix	1.5	1.5
CO.RT	CO (RSM)		Rect		

RR (W) : Width of AR effective
 RR (L) : Length of AR effective
 RROK : Allowed W=2.8,4,6,12,20
 GC (R) : GC in WN (R)
 RS (W) : Width of GR effective
 RS (L) : Length of GR effective
 RSOK : Allowed W=4,6,12,20
 WC : WN interacting AC

CO (RR) : CO on AR
 CO (RRW) : CO on AR (W>=4.0)
 CO (RRN) : CO on AR (W=2.8)
 ARN (S) : AR side fringe (W=2.8)
 ARW (S) : AR side fringe (W>=4.0)
 CO (C) : CO on WC
 CO (RS) : CO on GR
 GR (S) : GR side fringe
 CO (RSM) : CO (sized) & GR (S)

*Strike-through rules are not yet implemented

Cat-6: M1/V1/M2 related

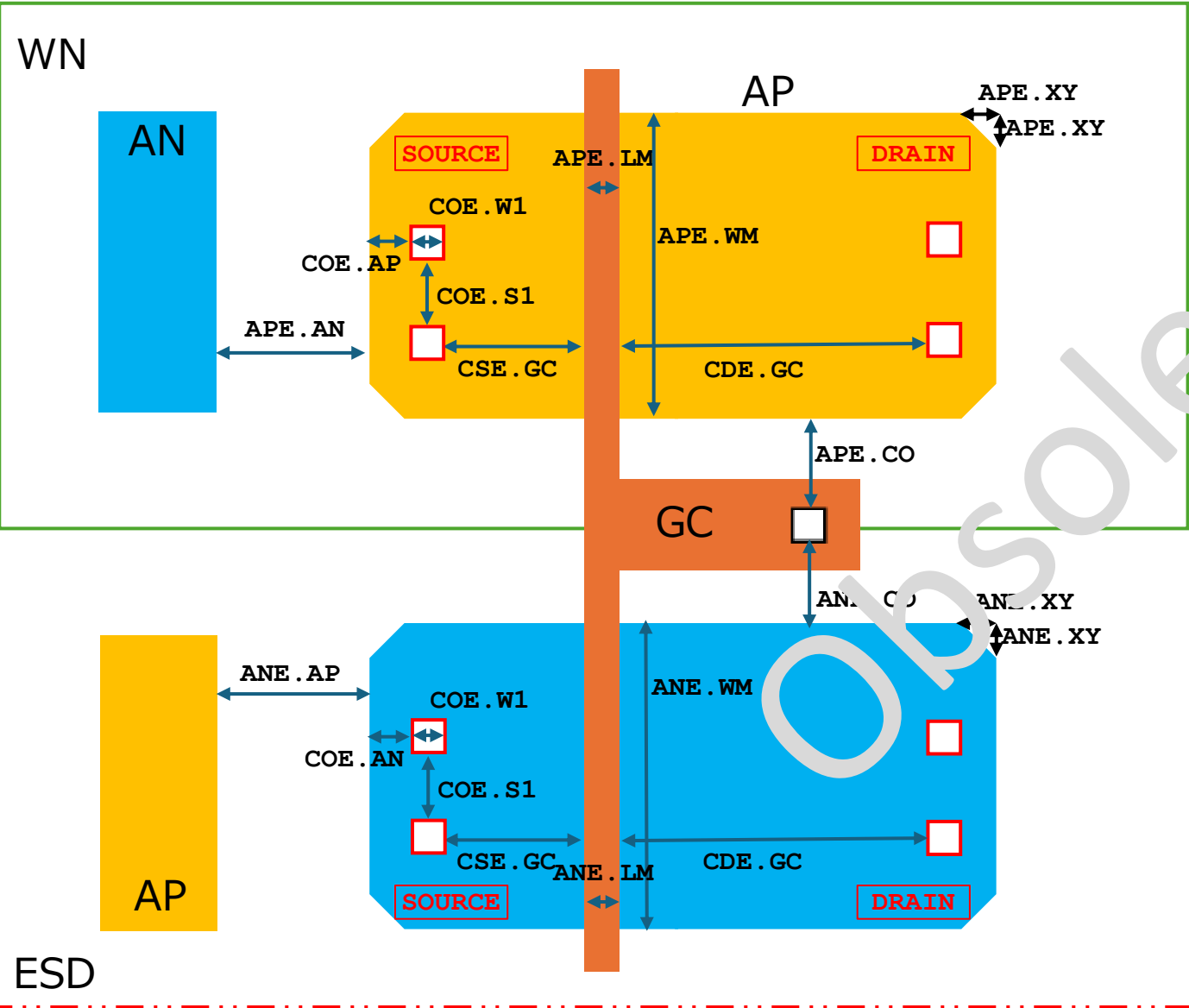


Rule	L1	L2	Func	MIN	MAX
M1.W1	M1		Wmin	1.8	
M1.W2	M1 (W)		Wmin	10.0	
M1.S1	M1	M1	Smin	1.4	
M1.SC	M1 (C)	M1 (C)	Smin/Smax	10.0	10.0
M1.CO	M1	CO	Fmin	0.8	
M1.CR	M1	CO (C)	Fmin	1.2	
M1.CC	M1	CO (CC)	Fmin	1.3	
M1.CL	M1	CO (L)	Fmin	1.2	
M1.SW	M1 (W)	M1	Smin	2.0	
V1.W1	V1		Wfix	1.4	1.4
V1.S1	V1	V1	Smin	1.5	
V1.M1	V1	M1	Emin	1.0	
V1.GC	V1	GC	Smin	1.2	
V1.CO	V1	CO	Smin	1.0	
V1.CL	V1	CO (L)	Smin	1.4	
M2.W1	M2		Wmin	3.0	
M2.S1	M2	M2	Smin	2.0	
M2.V1	M2	V1	Fmin	1.0	

M1(W) : Wide M1 (W>=10.0)
WN(C) : WN covering AC
M1(C) : M1 interacting (AN & WN(C))
CO(C) : CO on M1(C) (W=1.2um)
CO(CC) : CO on AC (W=1.2um)
CO(L) : CO large contact (Area>=1.5um²)

*Strike-through rules are not yet implemented

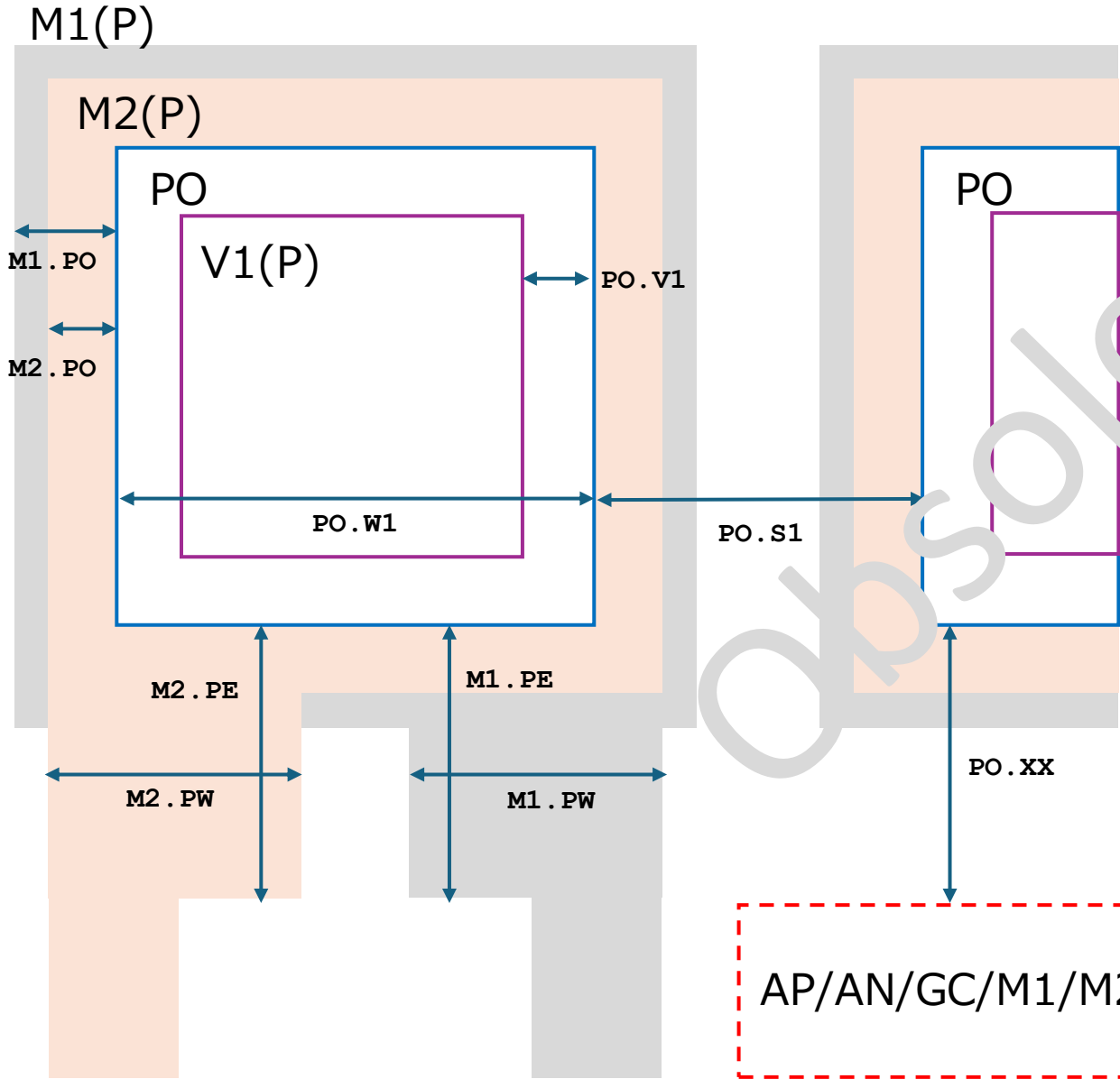
Cat-7: ESD MOS transistor related



Rule	L1	L2	Func	MIN	MAX
APE.WM	PMOSE	APE-GC	Wmin/max	11.0	60.0
APE.LM	PMOSE	GC-ANE	Lmin/max	2.0	30.0
APE.AN	APE	AN	Smin	10.0	
APE.XY	APE	Bevel	XYmin	1.4	
APE.CO	APE	CO	Smin	2.0	
ANE.WM	NMOSE	ANE-GC	Wmin/max	11.0	60.0
ANE.LM	NMOSE	GC-ANE	Lmin/max	2.0	30.0
ANE.AP	ANE	AP	Smin	10.0	
ANE.XY	ANE	Bevel	XYmin	1.4	
ANE.CO	ANE	CO	Smin	2.0	
COE.W1	CO (E)		Wfix	3.0	3.0
COE.S1	CO (E)	CO	Smin	1.6	
CDE.GC	CD (E)	GC	Smin	7.0	
CSE.GC	CS (E)	GC	Smin	3.0	
COE.AP	CO (E)	AP	Emin	4.0	
COE.AN	CO (E)	AN	Emin	4.0	

APE/ANE : AP/AN on ESD
CO (E) : CO on ESD

Cat-8: PAD related



Rule	L1	L2	Func	MIN	MAX
PO.W1	PO		Wmin	70.0	
PO.S1	PO	PO	Smin	64.0	
M2.PO	M2 (P)	PO	Fmin	5.0	
M1.PO	M1 (P)	PO	Fmin	5.0	
PO.V1	PO	V1 (P)	Emin	10.0	
PO.AP	PO	AP	Smin	14.0	
PO.AN	PO	AN	Smin	14.0	
PO.GC	PO	GC	Smin	14.0	
PO.M1	PO	M1	Smin	14.0	
PO.M2	PO	M2	Smin	14.0	
M2.PE	M2		EXTL	19.0	
M2.PW	M2		EXTW	40.0	
M1.PE	M2		EXTL	19.0	
M1.PW	M2		EXTW	40.0	

**** Not yet implemented**

M1 (P) : M1 covering PO
M2 (P) : M2 covering PO
V1 (P) : V1 through M1 (P) -M2 (P)

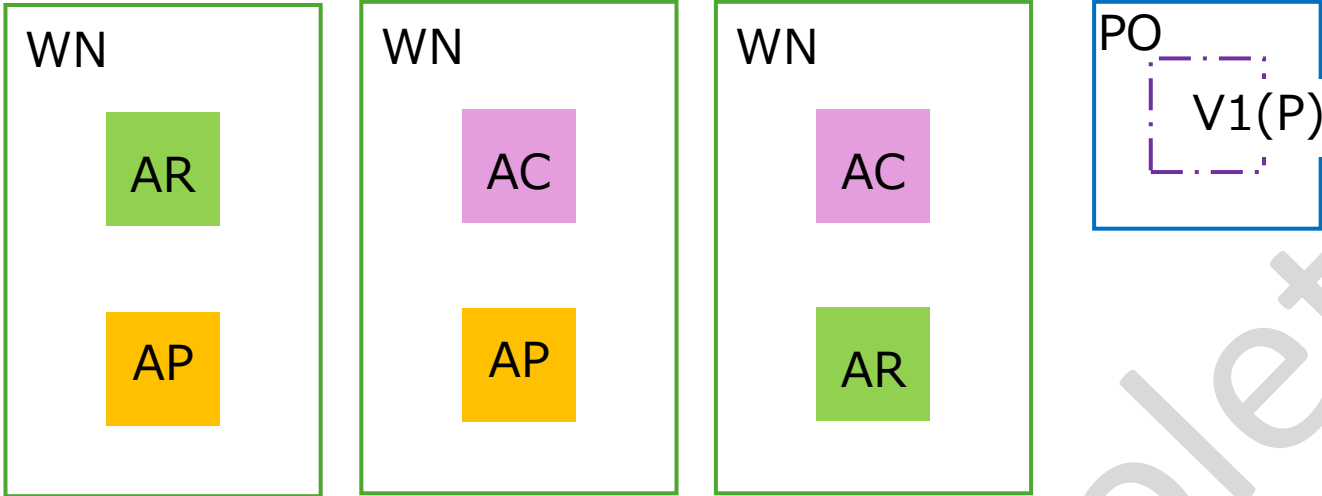
Obsolete slides

2026/06/11

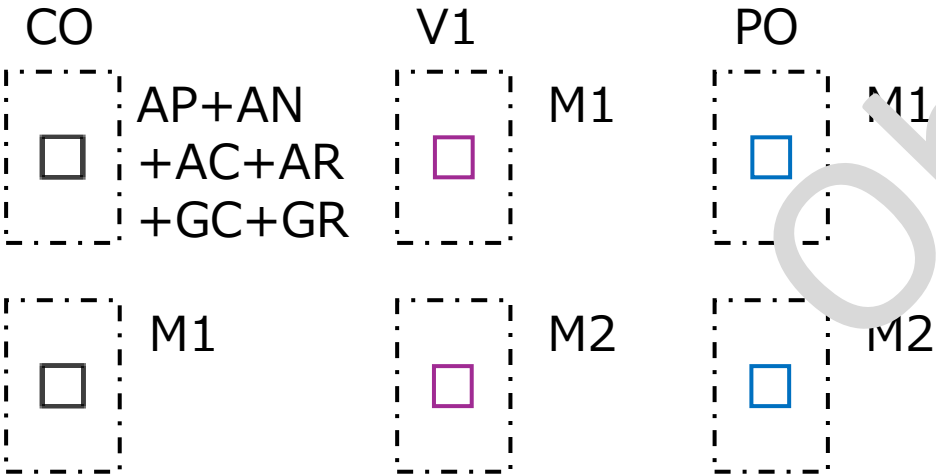
Reflected GitHub issues

<https://github.com/OpenSUSI/TR-1um/issues>

Cat-1: Prohibited



Rule	L1	L2	Func	MIN	MAX
WN.Z1	WN(R)	AP	None	0.0	
WN.Z2	WN(C)	AP	None	0.0	
WN.Z3	WN(C)	AR	None	0.0	
CO.Z1	CO	AA+GC+GR	None	0.0	
CO.Z3	CO	M1	None	0.0	
V1.Z1	V1	M1	None	0.0	
V1.Z2	V1	M2	None	0.0	
PO.Z1	PO	M1	None	0.0	
PO.Z2	PO	M2	None	0.0	
PO.Z3	PO	V1(P)	Exist	0.0	

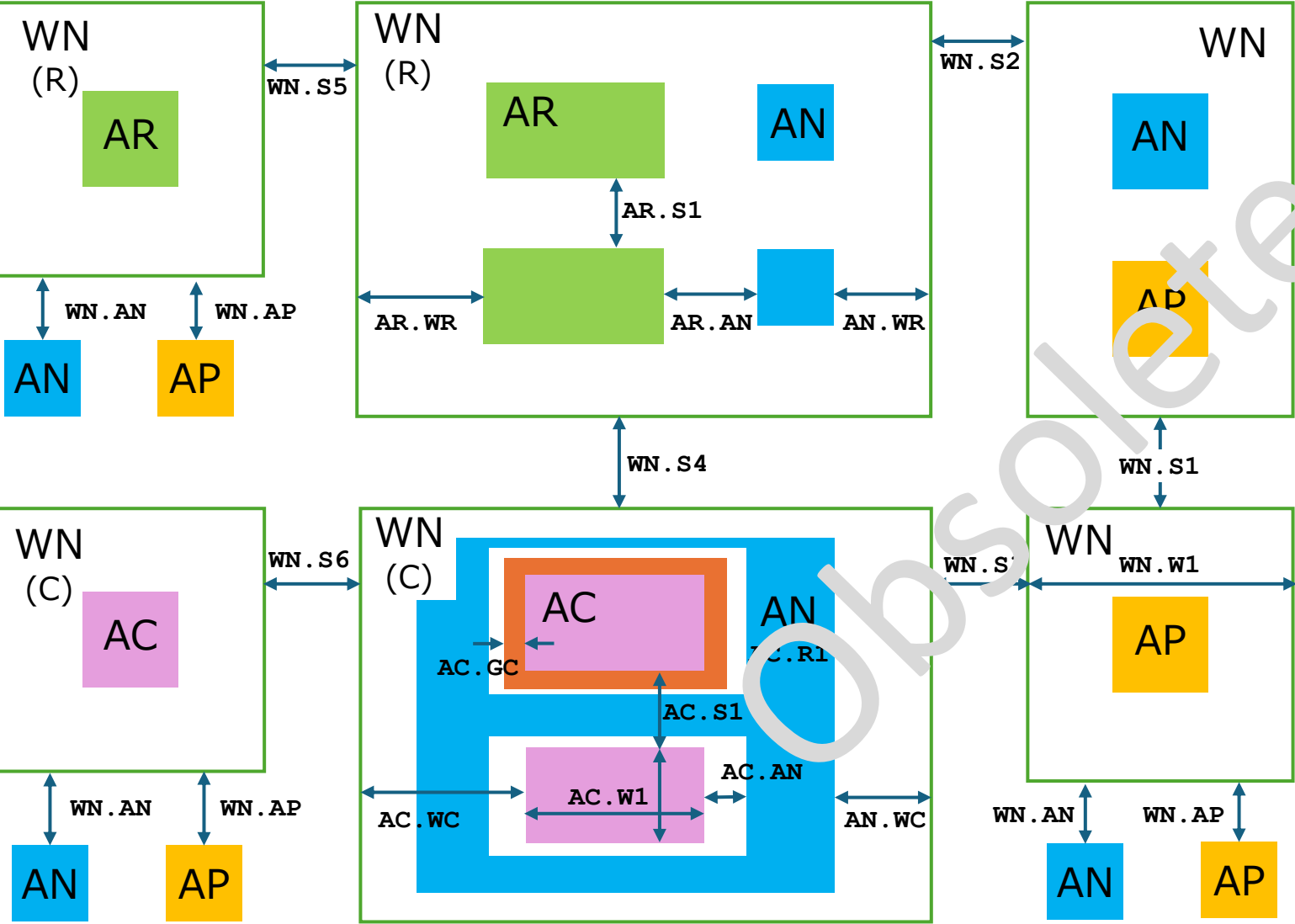


Function:

Wmin/max: L1 Width min/max
Smin/max: L1 Space min/max (L1=L2)
Smin/max: L1 Separate L2 min/max
Emin/max: L1 Enclosed L2 min/max
Fmin/max: L1 Enclosing L2 min/max
ECmin: MOS Endcap min
Wfix: L1 Windth equal
Sfix: Space equal
Donut: L1 Souuounded by L2
None: Not Exist

V1 (P) : V1 on PE

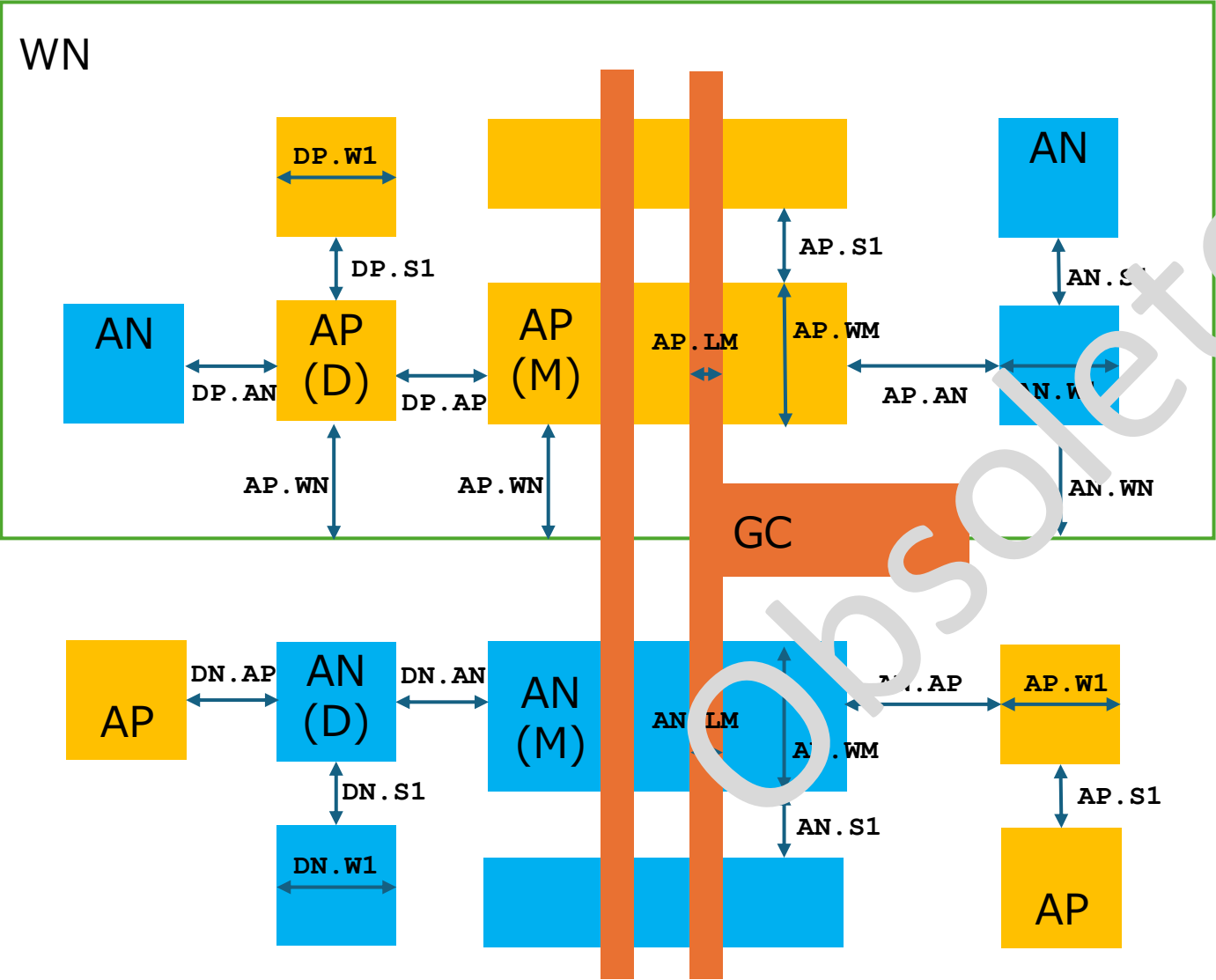
Cat-2: WN to AP/AN/AR/AC related



Rule	L1	L2	Func	MIN	MAX
WN.W1	WN		Wmin	8.0	
WN.S1	WN	WN	Smin	12.0	
WN.S2	WN	WN(R)	Smin	9.5	
WN.S3	WN	WN(C)	Smin	12.0	
WN.S4	WN(R)	WN(C)	Smin	9.5	
WN.S5	WN(R)	WN(R)	Smin	8.0	
WN.S6	WN(C)	WN(C)	Smin	12.0	
WN.AP	WN	AP	Smin	5.0	
WN.AN	WN	AN	Smin	10.0	
AR.S1	AR	AR	Smin	4.0	
AR.WR	AR	WN(R)	Emin	10.0	
AN.WR	AN	WN(R)	Emin	5.0	
AR.AN	AR	AN	Smin	4.0	
AC.W1	AC		Wmin/max	28.5	120.0
AC.S1	AC	AC	Smin	6.4	
AC.WC	AC	WN(C)	Emin	8.0	
AN.WC	AN	WN(C)	Emin	3.0	
AC.AN	AC	AN	Sfix	2.8	2.8
AC.GC	AC	GC	Efix	1.4	1.4
AC.R1	AC	AN(C)	Donut	0.0	

WN(R) : WN overlapped AR
WN(C) : WN overlapped AC
AN(C) : AN in WN(C)

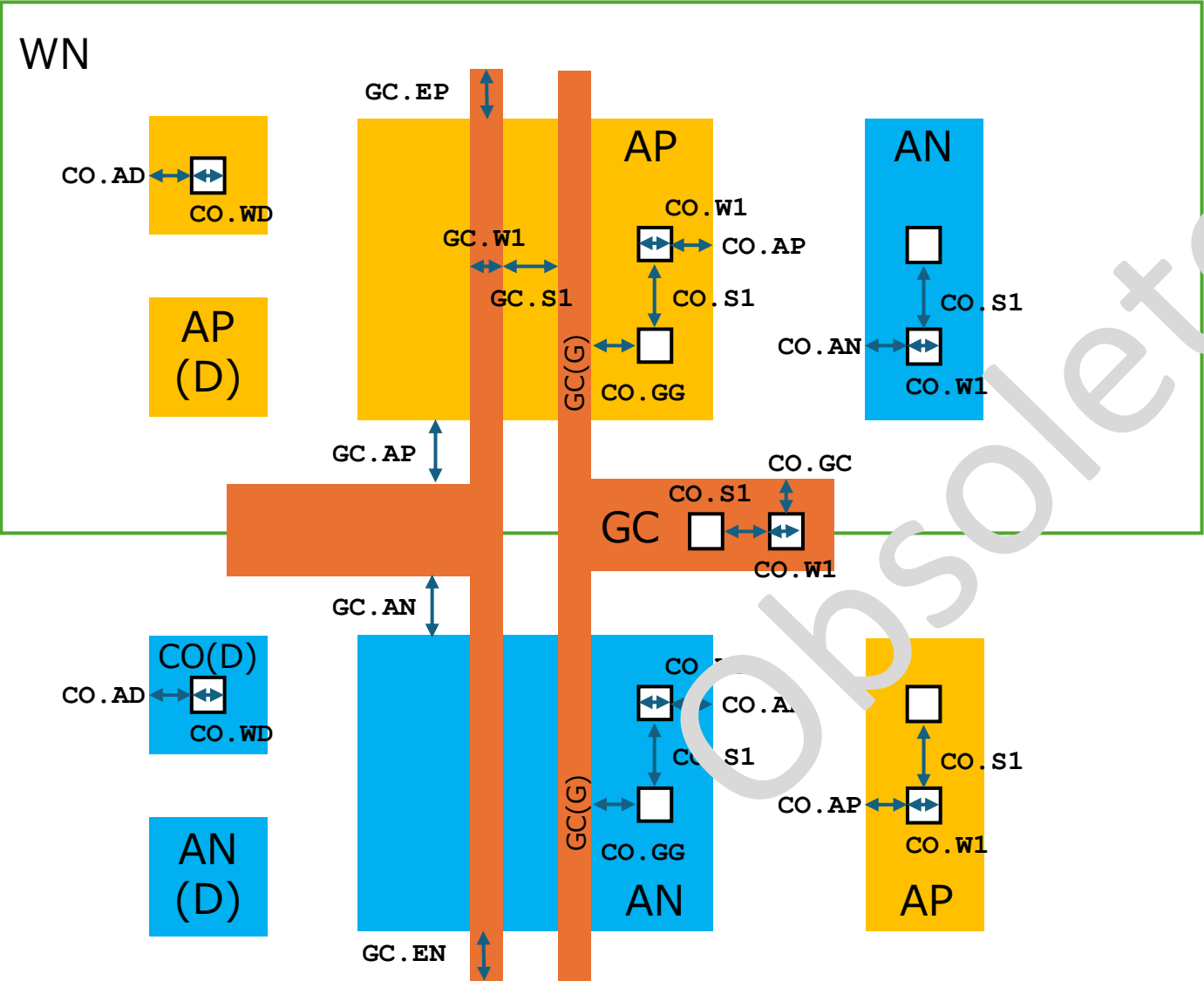
Cat-3: MOS transistor related



Rule	L1	L2	Func	MIN	MAX
AP.W1	AP		Wmin	1.4	
AP.S1	AP	AP	Smin	1.4	
AP.AN	AP	AN	Smin	2.8	
AP.WM	PMOS	AP-GC	Wmin/max	3.4	60.0
AP.LM	PMOS	GC-AN	Lmin/max	1.0	30.0
AP.WN	AP	WN	Emin	7.0	
DP.W1	DP		Wmin	2.8	
DP.S1	DP	DP	Smin	1.4	
DP.AP	DP	AP(M)	Smin	2.8	
DP.AN	DP	AN	Smin	2.8	
AN.W1	AN		Wmin	1.4	
AN.S1	AN	AN	Smin	1.4	
AN.AP	AN	AP	Smin	2.8	
AN.WM	NMOS	AN-GC	Wmin/max	3.4	60.0
AN.LM	NMOS	GC-AN	Lmin/max	1.0	30.0
AN.WN	AN	WN(M)	Emin	5.0	
DN.W1	DN		Wmin	2.8	
DN.S1	DN	DN	Smin	1.4	
DN.AP	DN	AP	Smin	2.8	
DN.AN	DN	AN(M)	Smin	2.8	

AN (M) : AN-DN
AP (M) : AP-DP
WN (M) : WN overlapped AN/AP

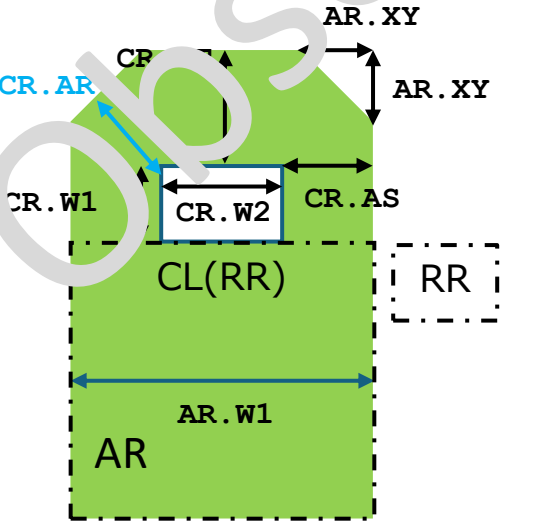
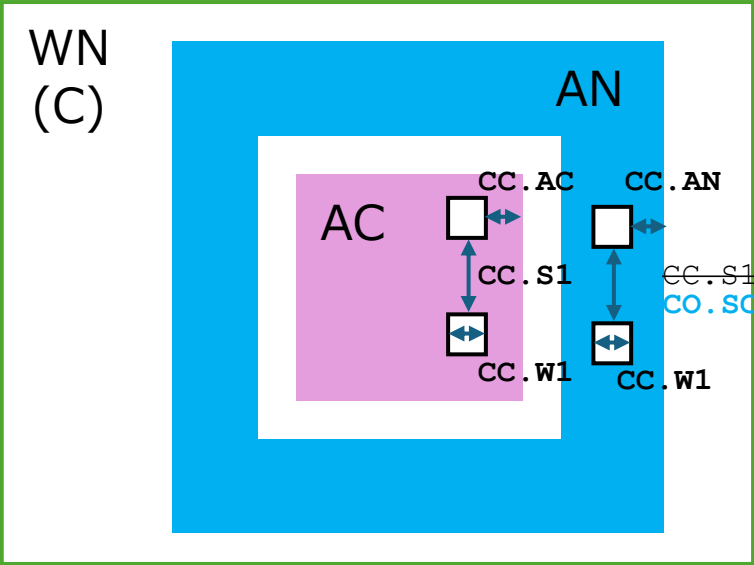
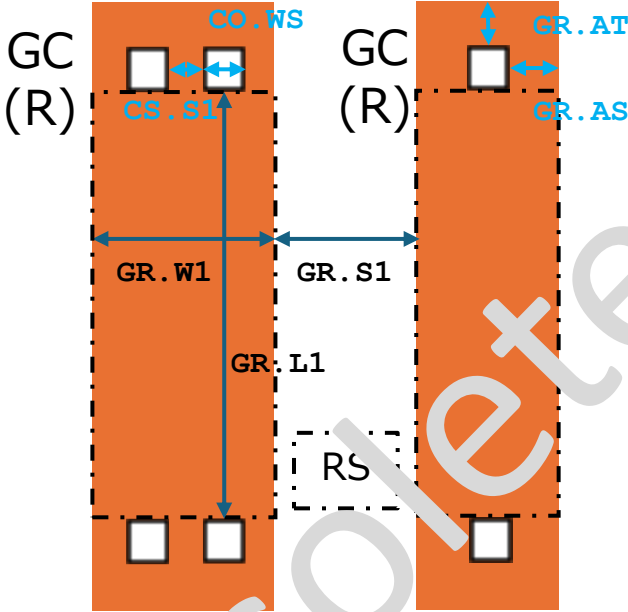
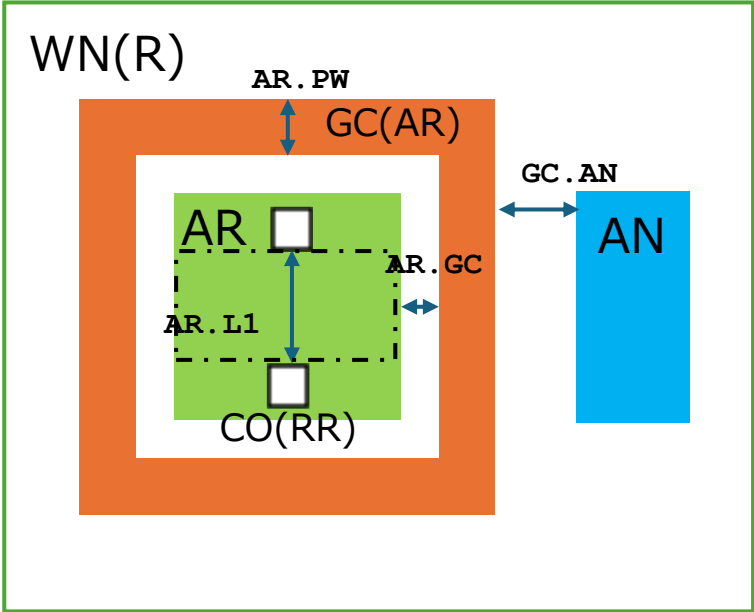
Cat-4: Contact and Poly related



Rule	L1	L2	Func	MIN	MAX
GC.W1	GC		Wmin	1.0	
GC.S1	GC	GC	Smin	1.2	
GC.AP	GC	AP	Smin	0.4	
GC.AN	GC	AN	Smin	0.4	
GC.EP	PMOS	GC	ECmin	1.2	
GC.EN	NMOS	GC	ECmin	1.2	
CO.W1	CO		Wfix	1.0	1.0
CO.S1	CO	CO	Smin	1.0	
CO.WD	CO(D)		Wfix	1.2	1.2
CO.AD	CO(D)	AP+AN	Emin	1.2	
CO.AP	CO	AP	Emin	0.8	
CO.AN	CO	AN	Emin	0.8	
CO.GG	CO	GC(G)	Smin	1.0	
CO.GC	CO	GC+GR	Emin	0.8	

AP(D) : AP not interacted GC
AN(D) : AN not interacted GC
CO(D) : CO on AP(D) /AN(D)
GC(G) : GC for PMOS/NMOS

Cat-5: Resistor(AR/RS) and Capacitor(AC) related



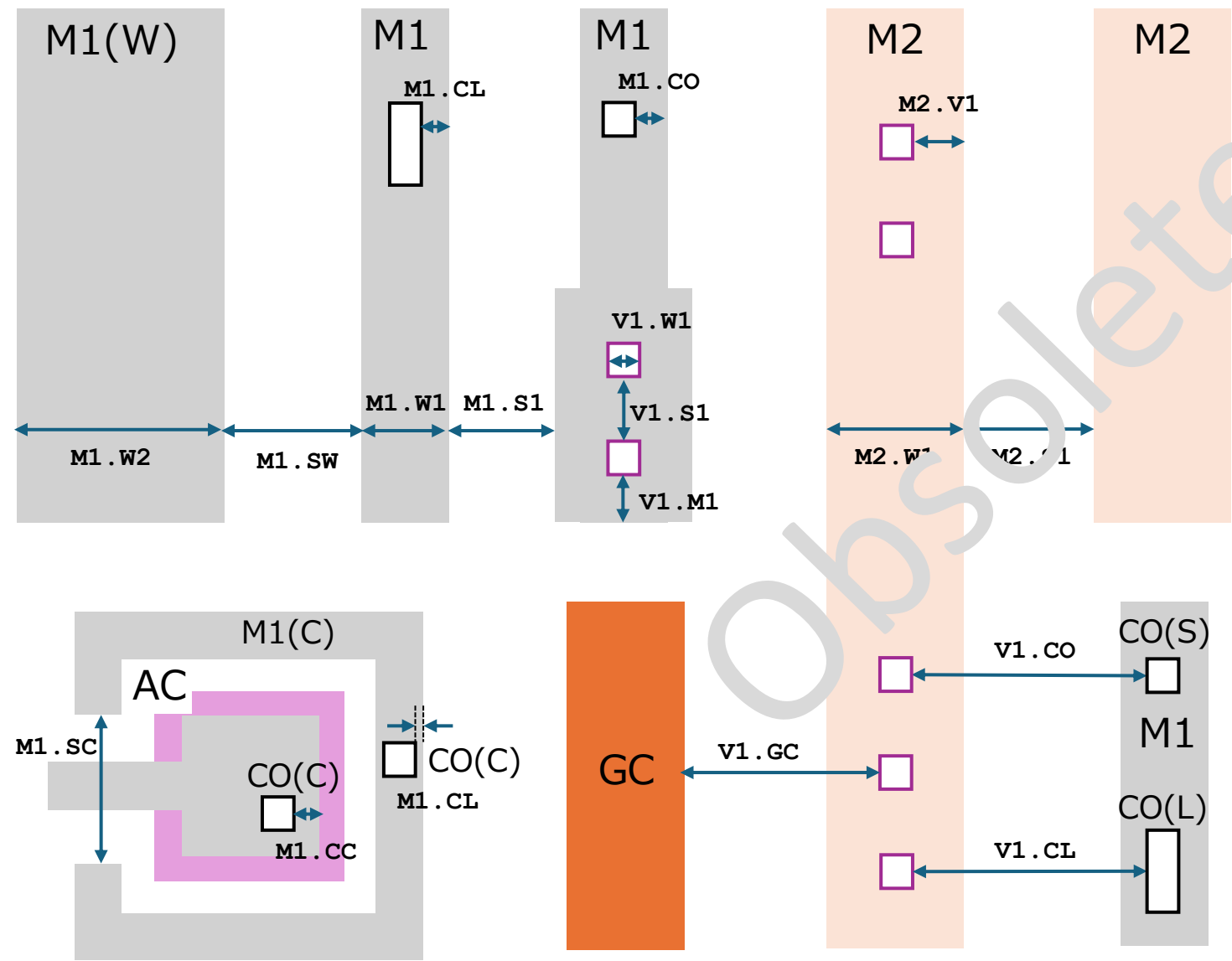
R.Effective region

Rule	L1	L2	Func	MIN	MAX
AR.W1	RR	RR(L)	Wmin/max	2.8	20.0
AR.WF	AR	ARF	Require		
AR.L1	RR	RR(W)	Lmin/max	13.0	100.0
AR.GC	GC	AR	Smin	1.0	
AR.PW	GC(R)		Wmin	2.0	
AR.XY	AR	Bevel	XYmin	1.0	
GC.AN	GC(R)	AN	Smin	???	???
GC.R1	AR	GC(R)	Donut	0.0	
GC.R2	GC(R)	WN(R)	TieDown	0.0	
GR.W1	RS	RS(L)	Wmin/max	4.0	20.0
GR.L1	RS	RS(W)	Lmin/max	20.0	100.0
GR.S1	RS	RS	Smin	2.0	
CR.W1	CO(RR)		Wmin	1.0	
CR.W2	CL(RR)		Wmin/max	1.0	20.0
CR.AR	CO(RR)	AR	Emin	0.8	
CR.AT	CO(RR)	AR(T)	Emin	1.5	
CR.AS	CO(RR)	AR(S)	Efix	0.9@W=2.8	1.2@W=>4
CC.W1	CO(C)		Wfix	1.2	1.2
CC.S1/CO.SC	CO(C)	CO(C)	Sfix/Smin	1.2	1.2
CC.AC	CO(C)	AC	Emin	2.9	
CC.AN	CO(C)	AN	Emin	1.2	
CO.WS	CO(S)		Wfix	1.0	1.0
CS.S1	CO(RS)	CO(RS)	Sfix	1.0	1.0
GR.AT	CO(RS)	GR(T)	Efix	0.8	0.8
GR.AS	CO(RS)	GR(S)	Efix	1.5	1.5

GC(AR) : GC in WN(R)
RS(L) : Length of AR effective
RS(W) : Width of AR effective
CO(RR) : CO short side on AR
CL(RR) : CO long side on AR
AR(T) : AR top fringe
AR(S) : AR side fringe
AR.W1 : allow only 2.8,4,6,12,20
GR.W1 : allow only 4,6,12,20

CO(C) : CO on AC
CO(S) : Small(1um□) CO
CO(RS) : CO on GR
GR(T) : GR top fringe
GR(S) : GR side fringe

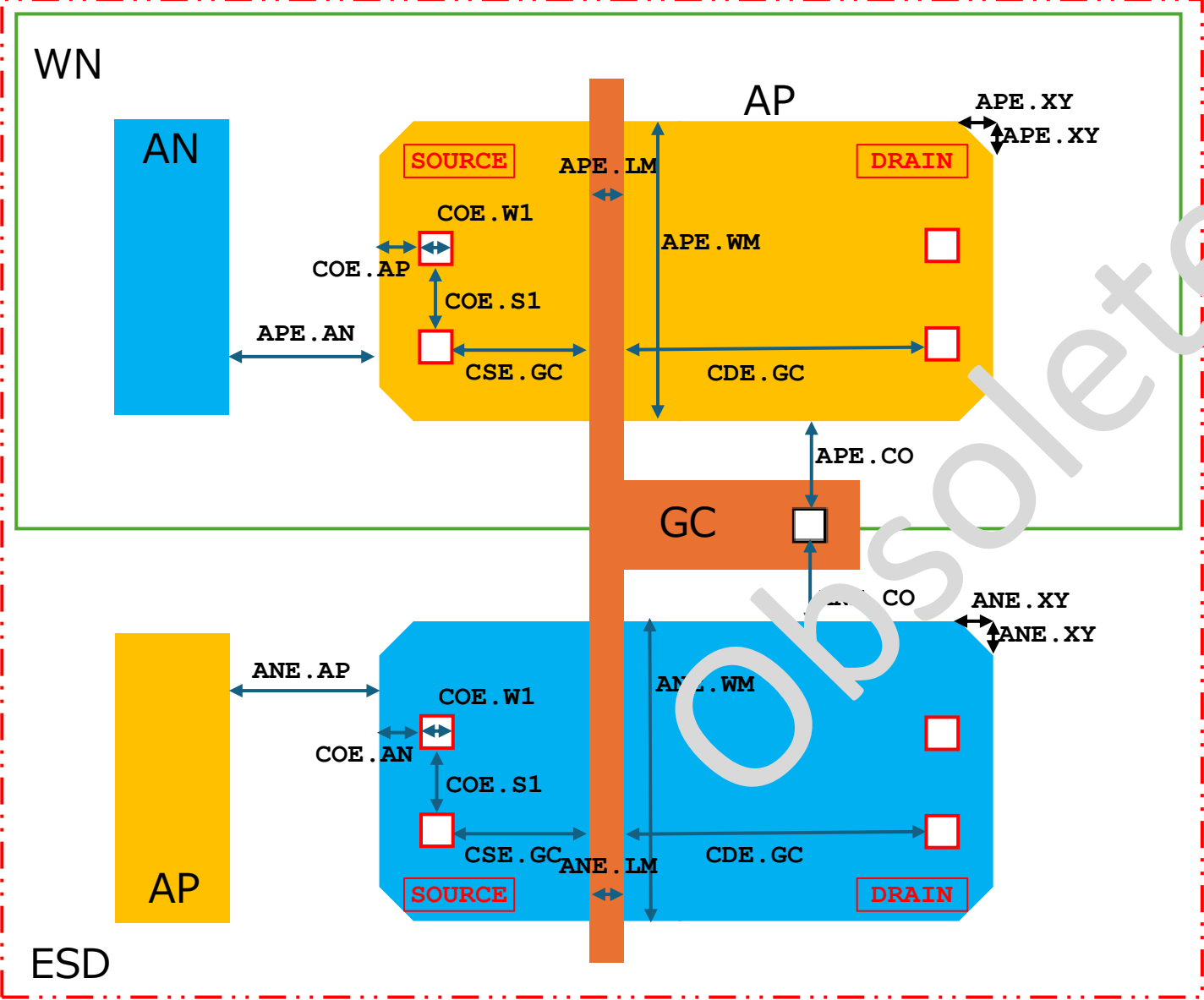
Cat-6: M1/V1/M2 related



Rule	L1	L2	Func	MIN	MAX
M1.W1	M1		Wmin/max	1.8	10.0
M1.W2	M1(W)		Wmin	10.0	
M1.S1	M1	M1	Smin	1.4	
M1.SC	M1(C)	M1(C)	Sfix	10.0	10.0
M1.CO	M1	CO(S)	Fmin	0.8	
M1.CC	M1	CO(C)	Fmin	1.3	
M1.CL	M1	CO(L)	Fmin	1.2	
M1.SW	M1(W)	M1	Smin	2.0	
V1.W1	V1		Wfix	1.4	1.4
V1.S1	V1	V1	Smin	1.5	
V1.M1	V1	M1	Emin	1.0	
V1.GC	V1	GC	Smin	1.2	
V1.CO	V1	CO	Smin	1.0	
V1.CL	V1	CO(L)	Smin	1.4	
M2.W1	M2		Wmin	3.0	
M2.S1	M2	M2	Smin	2.0	
M2.V1	M2	V1	Fmin	1.0	

M1 (C) : M1 in WN(C)
CO(C) : CO on AC
CO(S) : CO small contact
CO(L) : CO large contact (only on AR)
M1(W) : Wide M1

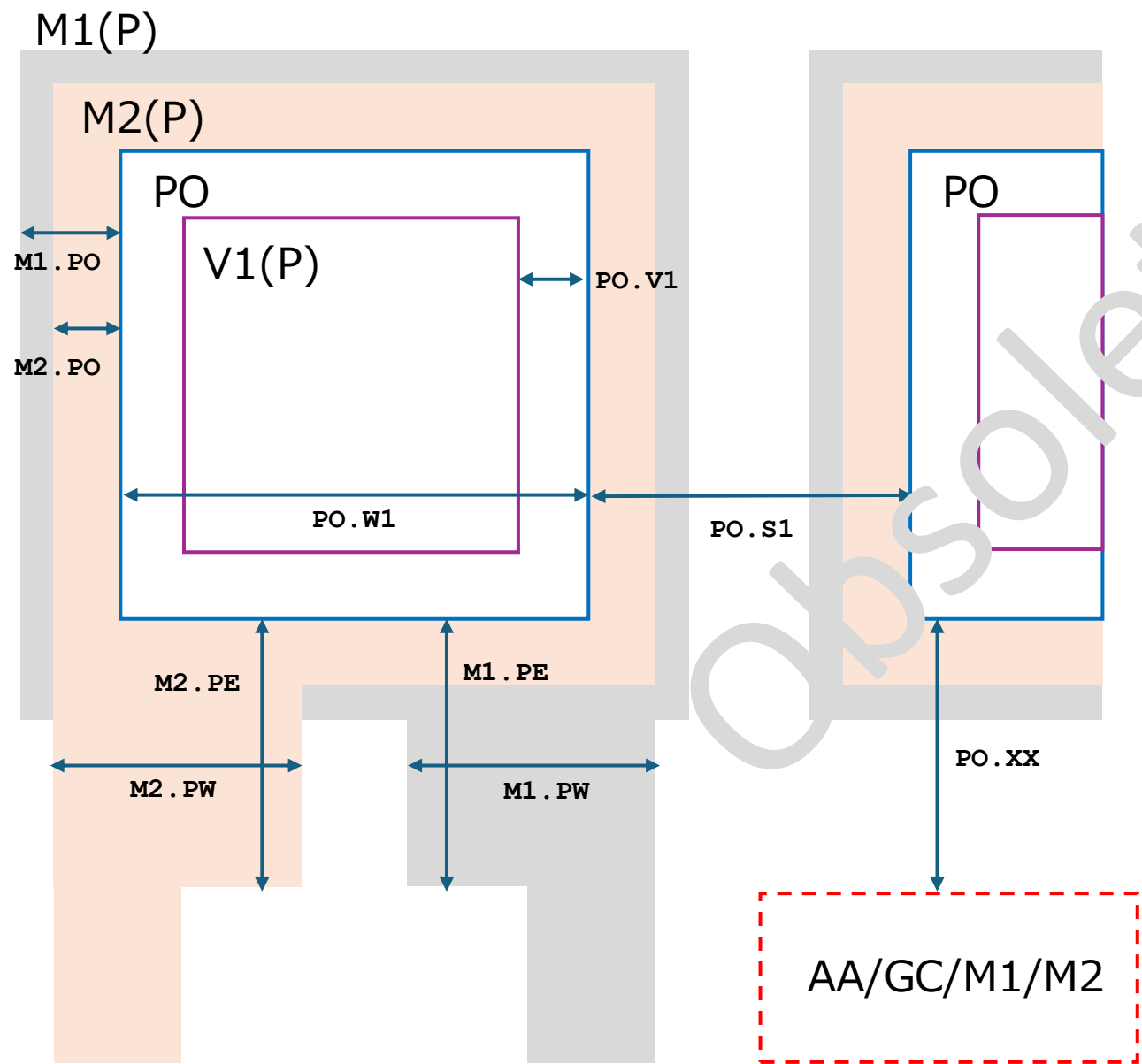
Cat-7: ESD MOS transistor related



Rule	L1	L2	Func	MIN	MAX
APE.WM	PMOSE	APE-GC	Wmin/max	11.0	60.0
APE.LM	PMOSE	GC-ANE	Lmin/max	2.0	30.0
APE.AN	APE	AN	Smin	10.0	
APE.XY	APE	Bevel	XYmin	1.4	
APE.CO	APE	CO	Smin	2.0	
ANE.WM	NMOSE	ANE-GC	Wmin/max	11.0	60.0
ANE.LM	NMOSE	GC-ANE	Lmin/max	2.0	30.0
ANE.AP	ANE	AP	Smin	10.0	
ANE.XY	ANE	Bevel	XYmin	1.4	
ANE.CO	ANE	CO	Smin	2.0	
COE.W1	CO(E)		Wfix	3.0	3.0
COE.S1	CO(E)	CO(E)	Smin	1.6	
CDE.GC	CD(E)	GC	Smin	7.0	
CSE.GC	CS(E)	GC	Smin	3.0	
COE.AP	CO(E)	AP	Emin	4.0	
COE.AN	CO(E)	AN	Emin	4.0	

APE/ANE) : AP/AN on ESD)
CO (E) : CO on ESD

Cat-8: PAD related



Rule	L1	L2	Func	MIN	MAX
PO.W1	PO		Wmin	70.0	
PO.S1	PO		Smin	64.0	
M2.PO	M2 (P)	PO	Fmin	5.0	
M1.PO	M1 (P)	PO	Fmin	5.0	
PO.V1	PO	V1 (P)	Emin	10.0	
PO.AP	PO	AP	Smin	14.0	
PO.AN	PO	AN	Smin	14.0	
PO.GC	PO	GC	Smin	14.0	
PO.M1	PO	M1	Smin	14.0	
PO.M2	PO	M2	Smin	14.0	
M2.PE	M2	Extention		19.0	
M2.PW	M2	Extention	Wmin	40.0	
M1.PE	M2	Extention		19.0	
M1.PW	M2	Extention	Wmin	40.0	

** Not yet implemented

M1 (P) : M1 on PO
M2 (P) : M2 on PO
V1 (P) : V1 on PO